

RMCPROFILE-SKILL · UNDERGRADUATE COURSE

Reverse Monte Carlo with RMCProfile

From the pair distribution function to a refined box in eleven lectures — every figure computed live in the companion notebooks

Why local structure?

A crystal structure is an **average**: one position per site, one displacement ellipsoid. Real materials disorder, tilt, hop and vibrate — and their properties live in that difference.

- Bragg peaks see the average unit cell; the diffuse scattering between them sees the deviations.
- Total scattering keeps both and Fourier-transforms them into the **pair distribution function**: how many atom pairs sit at each distance.
- Reverse Monte Carlo builds a box of thousands of atoms whose pair distribution matches the data — a picture of the disorder, not a parameter.
- RMCProfile is the program that does this for crystals, fitting $G(r)$, $F(Q)$ and the Bragg profile at once.

Eleven lectures

Each lecture is one run of slides: a picture first, the physics next, the equations last. Notebook sections are given for every lecture.

L0 Welcome — why local structure · §0–1

L1 Scattering and the PDF · §1–4

L2 Monte Carlo and Metropolis · §5–6

L3 Reverse Monte Carlo · §7–9

L4 Preparing a run · §10–14

L5 Fitting with RMCProfile · §15–18

L6 Constraints as physics · §23–27

L7 Corrections · §28–31

L8 Beyond neutrons · §19–22, §32–34

L9 Reading a configuration · §35–38

L10 Contributing to RMCProfile · §39–40

How each lecture is built

The deck is flat and linear: $\rightarrow$ is the only key you need. Every lecture opens with a divider and runs from its simplest slide to its most mathematical.

- **intro** slides: one figure, plain language, at most the symbols r and Q .
- **core** slides: the physics and how the toolkit or RMCProfile computes it (which function, which keyword).
- math slides: the formulas behind the figures, with the derivation sketched in the lecturer notes.
- To run the notebooks: create the `rmcprofile` environment (`environment-rmcprofile.yml`), open `chapters/README.md`, start with chapter 0; every chapter ends with exercises and worked solutions.
- Speaker view: `s`. Next lecture: `Shift+→`. Overview: `Esc`.

The one function of the course

Everything that follows is a way to compute, measure or fit Keen's total pair distribution function of a box of atoms.

PARTIAL PAIR DISTRIBUTION

$$g_{ij}(r) = n_{ij}(r) / (4\pi r^2 \rho c_j \Delta r)$$

KEEN'S G(R), IN BARN

$$G(r) = \sum_{ij} c_i c_j b_i b_j [g_{ij}(r) - 1]$$

ITS EXACT LIMIT

$$G(r \rightarrow 0) = - (\sum_i c_i b_i)^2$$

- $n_{ij}(r)$: pairs of type i – j in the shell at r ; ρ : number density; c : concentrations; b : neutron scattering lengths.
- The limit is a checkable number: for SF₆ the notebooks' weights give -0.2759 barn, the value at which the package's own $G(r)$ data file starts.
- RMCProfile's grid is $r_k = k \cdot \Delta r$ with bins centred on r_k ; on it the notebooks' partials equal the program's to 2.8×10^{-4} .

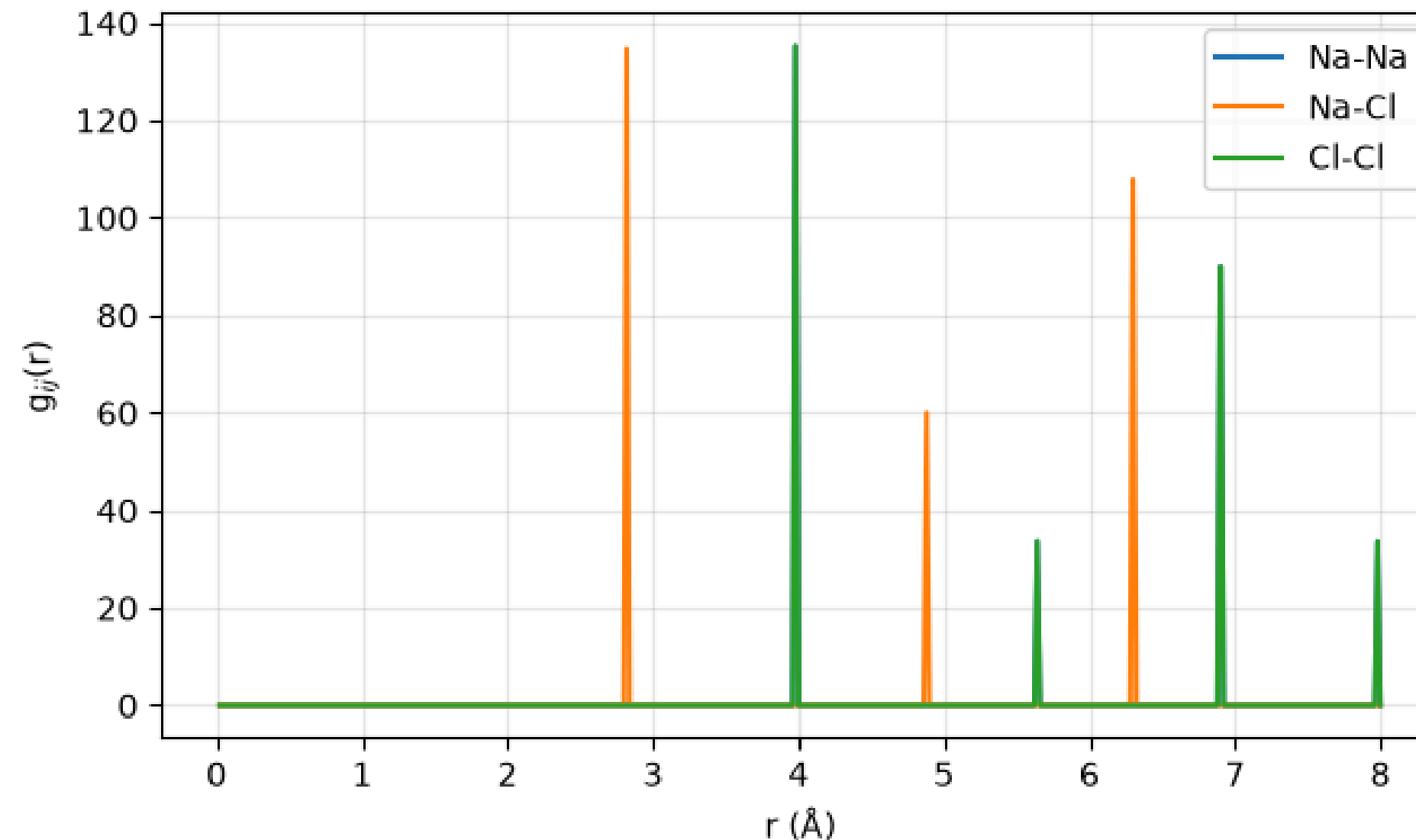
Scattering and the PDF

Partial pair distribution functions, thermal broadening, Keen's $G(r)$, $D(r)$ and $T(r)$ with their exact limits, and the Fourier route to $F(Q)$.

- 1 Counting pairs at each distance
- 2 Thermal motion broadens the peaks
- 3 Keen's total functions: $G(r)$, $D(r)$, $T(r)$
- 4 Into reciprocal space: $F(Q)$ and the box's ringing
- 5 The transforms

Counting pairs at each distance

Take a box of atoms, measure every pair distance, sort them into bins: the partial pair distribution functions. On an ideal NaCl lattice they are spikes at the shell distances.

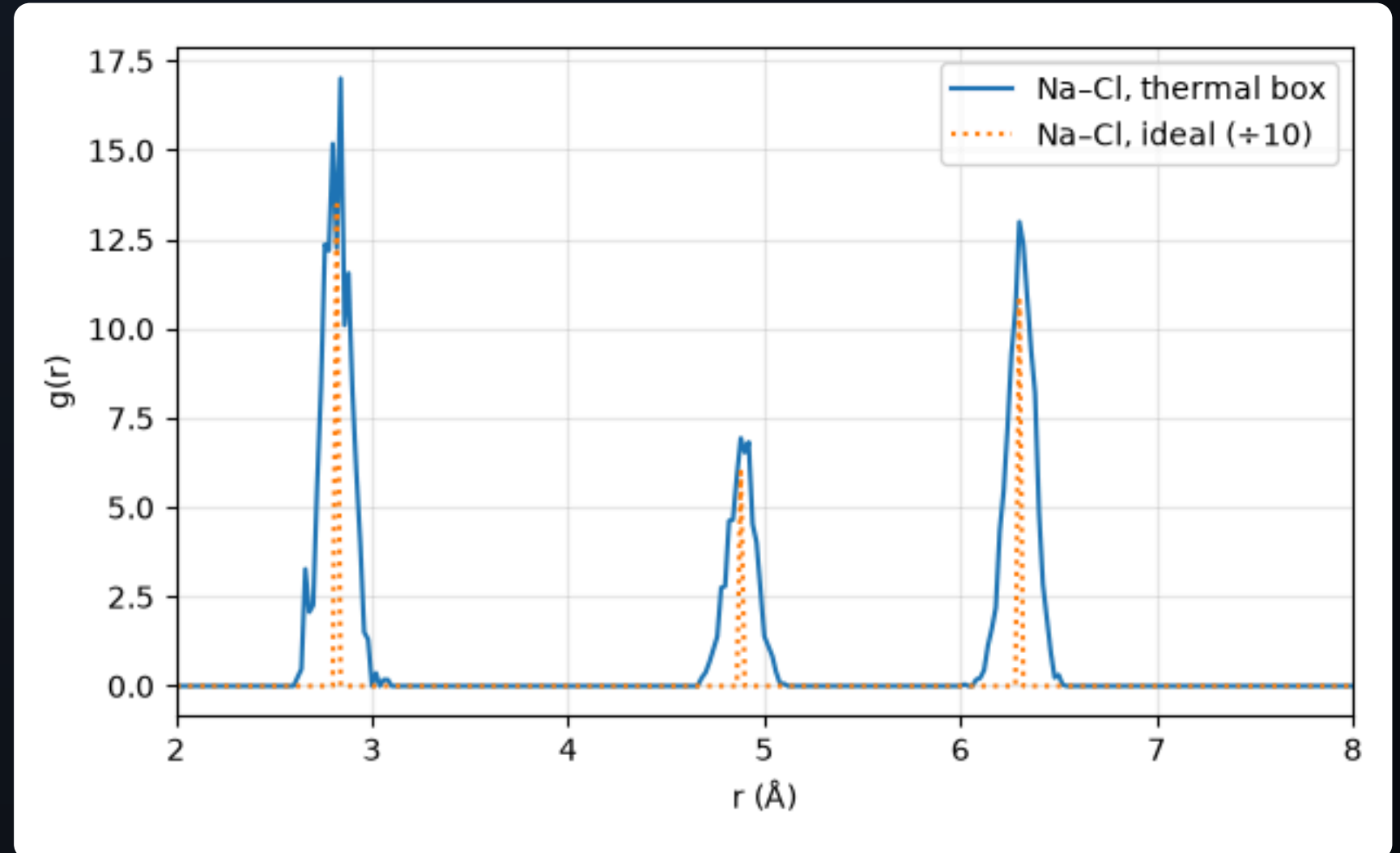


Partial pair distribution functions of the ideal NaCl lattice: every interatomic distance is a delta function on RMCProfile's r grid. The first Na-Cl shell sits at $a/2 = 2.82$ Å, the first Na-Na and Cl-Cl shells at $a/\sqrt{2} = 3.99$ Å. — Figure 1 · notebook §1 · chapter 1, cell 3

Thermal motion broadens the peaks

Displace every atom by a Gaussian of 0.05 Å and the spikes become peaks whose width is the pair's relative displacement — this is what a real material's PDF looks like.

- The peak **area** is conserved: the shell still holds six neighbours.
- The width of a pair peak is $\sqrt{2}$ times a single atom's displacement when the two move independently — correlated motion narrows it.
- The notebooks' 'truth box' for every later lecture is exactly this: an ideal lattice plus 0.05 Å Gaussian displacements, seed 0.

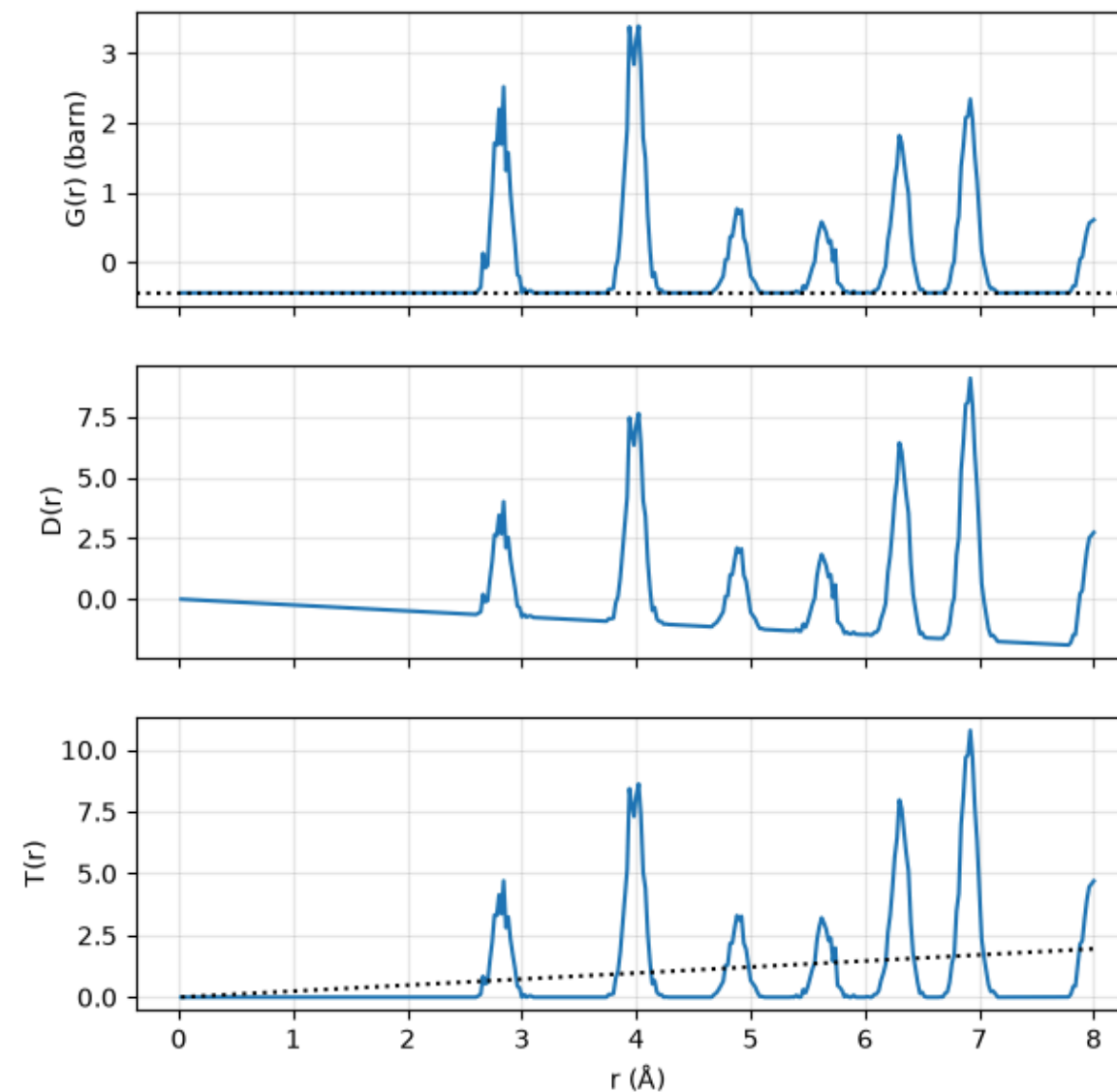


The Na-Cl partial of the thermally displaced box (0.05 Å per coordinate) against the ideal lattice's: the shells broaden into peaks of width ≈ 0.07 Å, the coordination numbers under them do not change.

– Figure 2 · notebook §2 · chapter 1, cell 5

Keen's total functions: $G(r)$, $D(r)$, $T(r)$

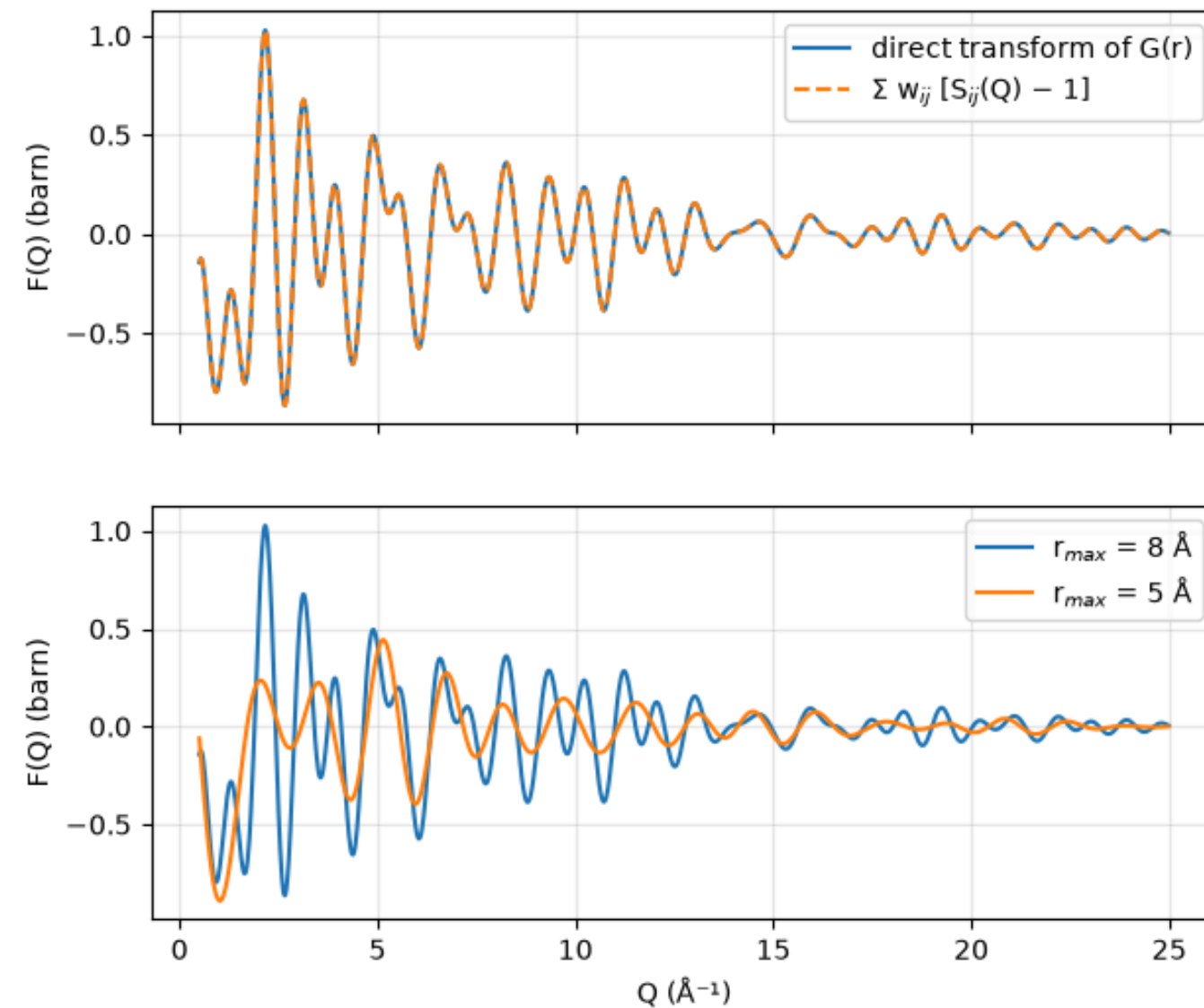
The partials are weighted by concentrations and scattering lengths into one measurable function. Three conventions differ by factors of r ; each has an exact limit the notebook checks.



Keen's three real-space functions for the thermal NaCl box. $G(r)$ starts at $-(\sum c b)^2$ (dotted), $D(r) = 4\pi r \rho_0 G(r)$ starts at zero with a negative slope, and $T(r)$ oscillates about the straight line $4\pi r \rho_0 (\sum c b)^2$ (dotted). — Figure 3 · notebook §3 · chapter 1, cell 7

Into reciprocal space: $F(Q)$ and the box's ringing

The same box in reciprocal space: Faber–Ziman partials $S_{ij}(Q) - 1$ by a sine transform of $g_{ij} - 1$, and their weighted sum $F(Q)$. A finite box rings.



Top: $F(Q)$ of the thermal box by the direct transform and through the Faber–Ziman partials — identical. Bottom: the same function from a 5 Å cut of the box: what the cut removes is the transform of $G(r)$ between 5 and 8 Å, a ripple whose Q -period is set by those shells — the reason RMCProfile convolves the data with the box function rather than pretending the model reaches infinity.

– Figure 4 · notebook §4 · chapter 1, cell 9

The transforms

Real space and reciprocal space are one function seen twice; the box makes the transform finite.

FABER-ZIMAN PARTIAL

$$S_{ij}(Q) - 1 = 4\pi \rho \int_0^{r_{max}} r^2 [g_{ij}(r) - 1] \sin(Qr)/(Qr) dr$$

TOTAL SCATTERING

$$F(Q) = \sum_{ij} c_i c_j b_i b_j [S_{ij}(Q) - 1]$$

THE OTHER CONVENTIONS

$$D(r) = 4\pi r \rho G(r), \quad T(r) = D(r) + 4\pi r \rho (\sum c b)^2$$

- The integral stops at $r_{max} \leq$ half the box edge: the minimum-image convention has no pairs beyond it.
- Truncation is a multiplication by a box in r , hence a convolution by a sinc in Q — the ringing of the figure.
- The notebooks' `fq_from_gr` and `total_fq_from_partials` are these two lines; the check is that they agree.

Monte Carlo and Metropolis

One atom moves, a histogram updates, a chi-squared changes; the Metropolis rule decides — the whole algorithm in one move.

- 1 One move, by hand
- 2 χ^2 and the acceptance rule
- 3 The engine in six lines: `rmclite`
- 4 Metropolis, with a temperature

One move, by hand

Pick one atom. Displace it by a random amount up to a maximum. Recount only the pairs that atom belongs to. Compare the new histogram with the target. Keep or undo.

- Nothing else changes: an **incremental** histogram update touches one row of distances, which is why a move costs microseconds.
- The notebooks' engine `rmcLite` checks that the incremental histogram stays bit-equal to a full rebuild.
- A closest-approach constraint can refuse the move before any histogram is touched.

χ^2 and the acceptance rule

The distance between the calculated and the target function is a sum of squared residuals over bins, each divided by an uncertainty σ . A move that lowers χ^2 is always kept; one that raises it is kept sometimes.

- The 'sometimes' is Metropolis: with probability $\exp(-\Delta\chi^2/2)$. It lets the box climb out of local minima and gives the box a temperature.
- σ is the knob that sets how far the data pull: a small σ makes every wiggle of the target — including its noise — a mandatory feature.
- Several data sets add their χ^2 terms with weights; constraints and potentials add terms of their own, exactly as RMCProfile's chi2 file lists them.

The engine in six lines: `rmclite`

The notebooks' own reverse Monte Carlo engine reuses the toolkit's grid and normalisation, so what it fits is exactly what RMCProfile tabulates.

```
import rmclite as rl

box = rl.Box.from_rmc6f(cfg)                # the average structure, from an rmc6f
targets = [rl.PartialTarget(lab, r, g_truth[lab], sigma=0.2) for lab in g_truth]
engine = rl.RmcLite(box, rmax=8.0, dr=0.02, targets=targets,
                    constraints=[rl.ClosestApproach({"Na-Na": 3.0, "Na-Cl": 2.2, "Cl-Cl": 3.0})],
                    max_move={"Na": 0.05, "Cl": 0.05}, seed=1)
engine.run(6000)                           # chi2 falls by more than 95 %
```

- `Box`: fractional coordinates and the lattice; `Histogram`: the incremental pair count on RMCProfile's grid.
- Targets: `PartialTarget`, `TotalGTarget`, `FqTarget`; constraints: `ClosestApproach`, `DistanceWindow`; a `BondPotential` adds $\Delta U/k_B T$ to the rule.
- About 1 900 moves per second on 64 atoms in pure numpy — RMCProfile does ten times more with a $G(r)$ data set, and the point of `rmclite` is to be read, not to be fast.

Metropolis, with a temperature

The acceptance rule is one inequality; a potential energy enters it on the same footing as the data.

CHI-SQUARED

$$\chi^2 = \sum_{data} \sum_k [f_{calc}(r_k) - f_{data}(r_k)]^2 / \sigma^2$$

THE DECISION

$$\Delta = \Delta\chi^2/2 + \Delta U/k_B T, \text{ accept if } \Delta \leq 0, \text{ else with probability } e^{-\Delta}$$

HARMONIC BOND TERM

$$U = 1/2 k (d - d_o)^2$$

- With no data at all the rule samples the Boltzmann distribution of U : rmlite's test lets a harmonic bond thermalise to $\langle U \rangle = 0.78 k_B T/k$, the equipartition check.
- RMCProfile's `POTENTIALS ::` block (STRETCH, TEMPERATURE) is this term with k in eV/Å² and T in kelvin.
- The factor 1/2 on $\Delta\chi^2$ is the convention of the original RMC papers (McGreevy & Pusztai 1988) and of RMCProfile.

Reverse Monte Carlo

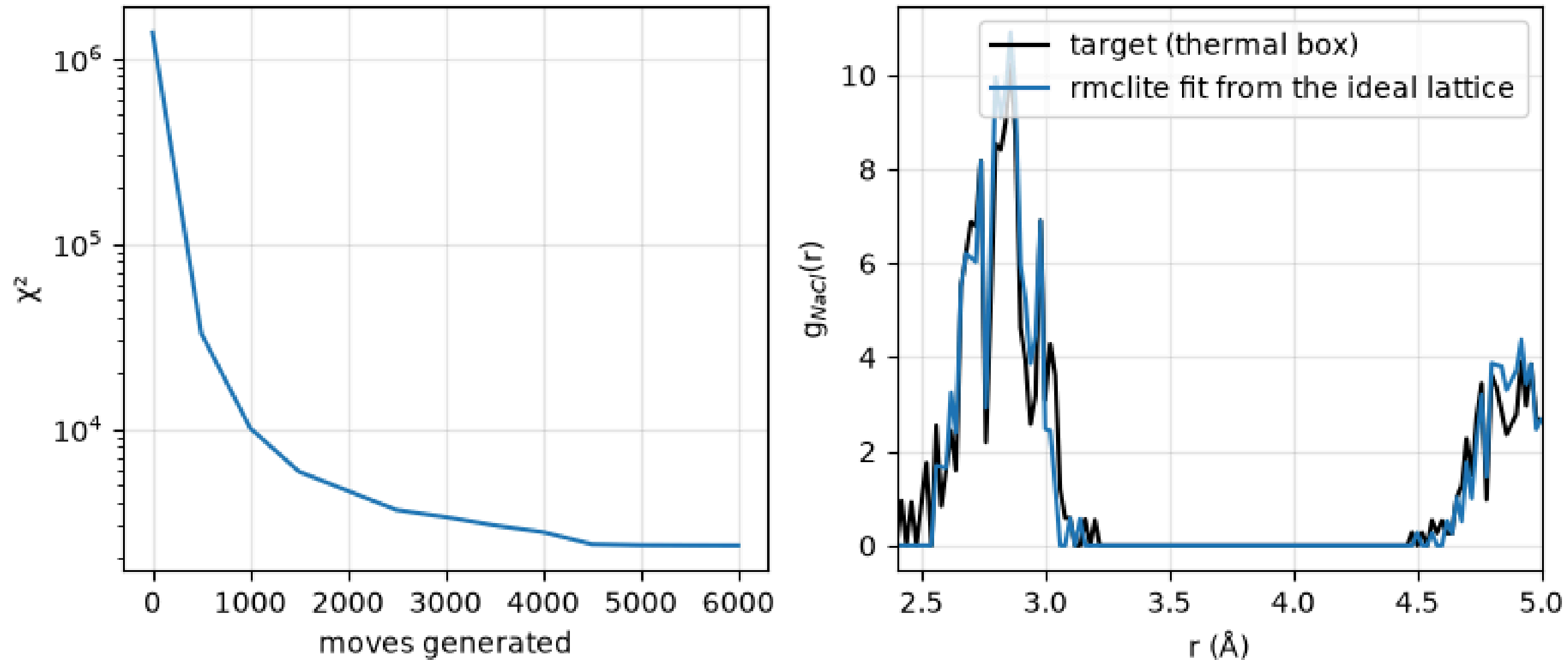
From the average structure to the data: what a fit reproduces (the pair distribution), what it does not (the coordinates), and why constraints are physics.

- 1 From the average structure to the data
- 2 What RMC does and does not recover

- 3 Constraints are physics
- 4 Why the coordinates are not recoverable

From the average structure to the data

Start rmc-lite on the ideal NaCl lattice with the thermal box's partials as targets: the spikes broaden into the target peaks and χ^2 falls by four orders of magnitude.



Left: χ^2 against generated moves for a fit from the ideal NaCl lattice toward the partials of a thermally displaced box. Right: the Na-Cl partial of the fit against its target after 6000 moves.

— Figure 5 · notebook §7 · chapter 2, cell 8

What RMC does and does not recover

The fit reproduces the **pair distribution**; it does not recover the **coordinates**. The residual against the target is as small as that of a second, independent thermal realisation — and the atoms are not where the truth put them.

- Chapter 2 measures it: the fitted box's partials deviate from the target no more than a fresh random realisation's; the fitted displacements are uncorrelated with the truth's.
- So read a refined box for its **distributions** — bond lengths, angles, coordination, displacement clouds (lecture 9) — never for atom 137.
- Starting point matters: start from the average structure; a random start converges to a different, equally good, box.
- Move size: 0.3 Å converges as well as 0.05 Å on NaCl; 0.01 Å is slow. The notebook's exercise has the numbers.

Constraints are physics

A pair distribution constrains distances, not chemistry. A closest approach, a distance window, a coordination number or a bond potential is physics the data cannot say and the fit needs.

- Without a closest approach two atoms can overlap and still fit a broad peak; with one, every move is checked before the histogram is touched.
- Chapter 2's constrained fit against the same targets converges to a box whose Na–Cl distances never fall below 2.2 Å; the unconstrained one wanders.
- RMCProfile carries the same ideas as keywords — `MINIMUM_DISTANCES`, `DISTANCE_WINDOW`, `POTENTIALS`, `FIXED_COORDINATION_CONSTRAINTS`, `BVS` — lecture 6 measures each on a refined box.

Why the coordinates are not recoverable

The data fix pair statistics; a permutation of displacements with the same pair statistics fits as well.

WHAT THE TARGET FIXES

$$g_{ij}(r) \propto \Sigma_{pairs} \delta(r - |r_a - r_b|)$$

A SYMMETRY OF THE FIT

$$\{u_a\} \rightarrow \{u_{\pi(a)}\} \text{ for any } \pi \text{ that preserves the shell statistics}$$

THE HONEST RESIDUAL

$$rms(g_{fit} - g_{target}) \approx rms(g_{realisation\ 2} - g_{target})$$

- The number of coordinates (3N) vastly exceeds the number of independent bins, and the bins are sums over all pairs.
- Chapter 2 measures the third line: the fit's residual against the target equals a second thermal realisation's within noise.
- The consequence for lecture 9: report distributions with the ensemble spread as the error bar.

Preparing a run

A unit cell becomes a supercell and an rmc6f file; the .dat keyword file and the checker; synthetic data with a known answer.

- 1 From a unit cell to a supercell
- 2 Folding back, averaging, exporting
- 3 The .dat file: keywords, blocks, flags
- 4 Check before you run
- 5 Synthetic data with a known answer

From a unit cell to a supercell

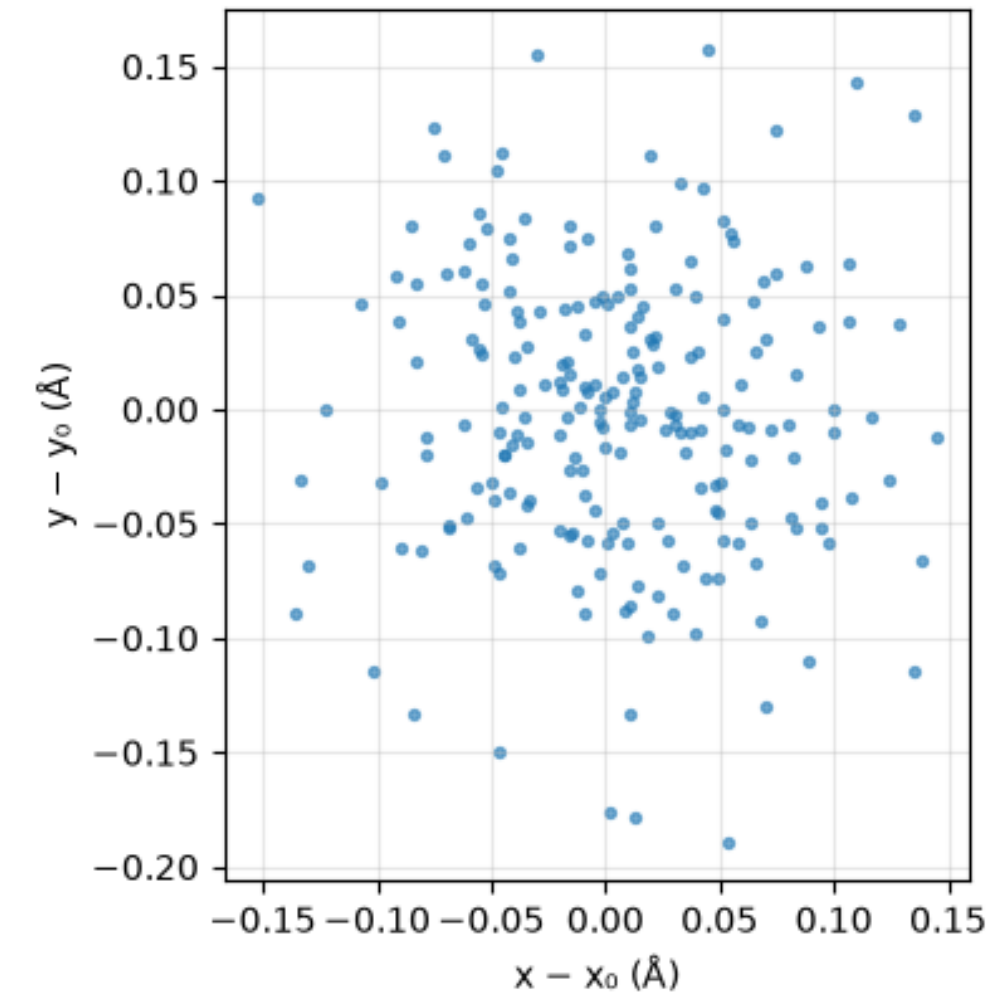
RMCProfile needs a box: a cell, a list of sites with fractional coordinates, and how many times to repeat it. The toolkit's `build_configuration` turns a crystallographic description into an rmc6f file.

- NaCl: $a = 5.64 \text{ \AA}$, four Na and four Cl sites; $3 \times 3 \times 3$ cells give 216 atoms in a $16.92 \text{ \AA}$ box.
- The rmc6f file carries the cell, the atom types in a fixed order, and each atom's fractional position with its site and cell indices — the program writes the same layout back.
- Bigger boxes give smoother partials and longer r_{maxi} ; the shipped SF_6 exercise uses 4 320 atoms.

Folding back, averaging, exporting

Fold every atom of the supercell into one unit cell and the displacement clouds of the sites appear; average the folded positions and the unit cell is back. Export to XYZ or CIF for any viewer.

- `fold_to_unit_cell`, `average_cell`, `export_xyz`, `export_cif` — all on the `rmc6f` object.
- The CIF's cell is the supercell (16.92 Å); the average cell is the unit cell (5.64 Å): chapter 9's exercise checks both.
- Fold-back is the bridge between an RMC box and a crystallographer's picture.



The 216 Sr atoms of the thermal SrTiO_3 box folded back into one unit cell and projected on the ab plane: a Gaussian cloud around the site whose width is the displacement amplitude put in (0.06 Å per coordinate).

– Figure 6 · notebook §12 · chapter 3, cell 7

The .dat file: keywords, blocks, flags

One text file drives a run: scalars, data blocks with subordinate keywords, and bare flags. The toolkit writes a complete synthetic set in one call and keeps the three keyword forms apart.

```
rt.write_input_set("nacl", workdir, cfg, gr=(r, G_data),
                  min_dist={"Na-Na": 3.0, "Na-Cl": 2.2, "Cl-Cl": 3.0},
                  max_move=0.05, time_limit_min=0.5)

# the file it writes (excerpt)
#  MINIMUM_DISTANCES :: 3.0 2.2 3.0
#  NEUTRON_REAL_SPACE_DATA :: 1
#    > FILENAME :: nacl_gr.dat
#    > DATA_TYPE :: G(r)
#    > CONVOLVE ::          <- a keyword with an empty value
#    > NO_FITTED_OFFSET    <- a bare flag
#  IGNORE_HISTORY_FILE ::
```

- `> KEY :: value`, `> KEY ::` and `> KEY` are three different things to the program; a writer that drops the `::` of `CONVOLVE` silently disables the convolution (finding P-16).
- `IGNORE_HISTORY_FILE ::` is on by default: a `.his6f` left by a zero-move pass poisons the next run (P-15).
- `MINIMUM_DISTANCES` lists pairs in the order AA AB AC BB BC CC; the checker counts them.

Check before you run

RMCPProfile stops on a missing file with exit code 0 and waits forever for a missing .poly file. The toolkit's checker reports the mistakes the manual warns about — and the ones it does not — before the binary starts.

code	level	what it catches
no-configuration	ERROR	no rmc6f / his6f / cfg — the program would stop with exit code 0 (P-11)
minimum-distances-count	ERROR	wrong number of values for the pairs AA AB AC BB BC CC
poly-file-missing	ERROR	POLYHEDRAL_RESTRAINT without its .poly — the program waits indefinitely (P-12)
bulk-rho-missing	ERROR	PARTICLE_RADIUS without BULK_RHO — the program stops (P-18)
filename-case	WARN	a data file that exists only with different letter case — fine on Windows, a stop on Linux (P-10)
history-file	WARN	a .his6f will be read instead of the rmc6f; one from a zero-move pass makes the run crawl (P-15)
stale-neighbour-files	WARN	a .neigh from a previous configuration

Synthetic data with a known answer

The course's experiments are built so that the truth is known and the fit's success is measurable.

TRUTH BOX

$$x_a = x_a^{ideal} + u_a, \quad u_a \sim N(0, 0.05 \text{ \AA}) \text{ per Cartesian component, seed 0}$$

DATA

$$G_{data}(r_k) = G_{truth}(r_k) + N(0, 0.003 \text{ barn})$$

GRID

$$r_k = k \cdot \Delta r, \quad \Delta r = 0.02 \text{ \AA}, \quad r_{max} = 8 \text{ \AA} < L/2 = 8.46 \text{ \AA}$$

- The data file is written in RMCProfile's two-line-header layout; the program reads it back exactly (the experiment column returns our numbers to 10^{-4}).
- Thresholds in the notebooks are stated relative to the noise — 'below ten times the noise' — never tuned to pass.
- Every later lecture reuses this box; when a lecture evaluates a keyword it uses the truth box itself as the model, so only the keyword can differ.

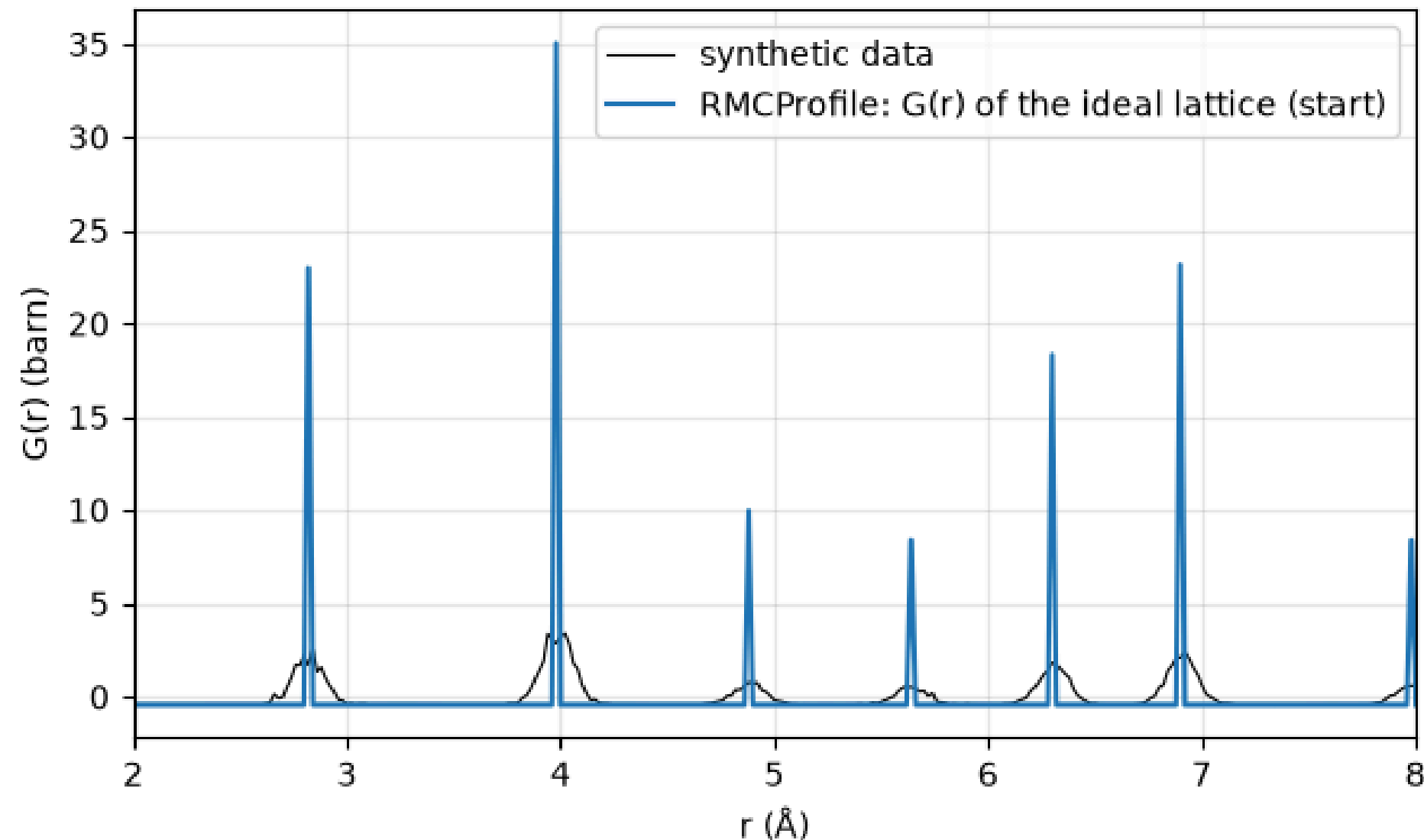
Fitting with RMCProfile

RMCProfile computes what the toolkit computes; a real refinement of synthetic neutron data; CONVOLVE and the finite box; the same data through rmclite.

- 1 One pass: RMCProfile computes what we compute
- 2 The refinement
- 3 CONVOLVE :: and the finite box
- 4 The same data through rmclite
- 5 The box function

One pass: RMCProfile computes what we compute

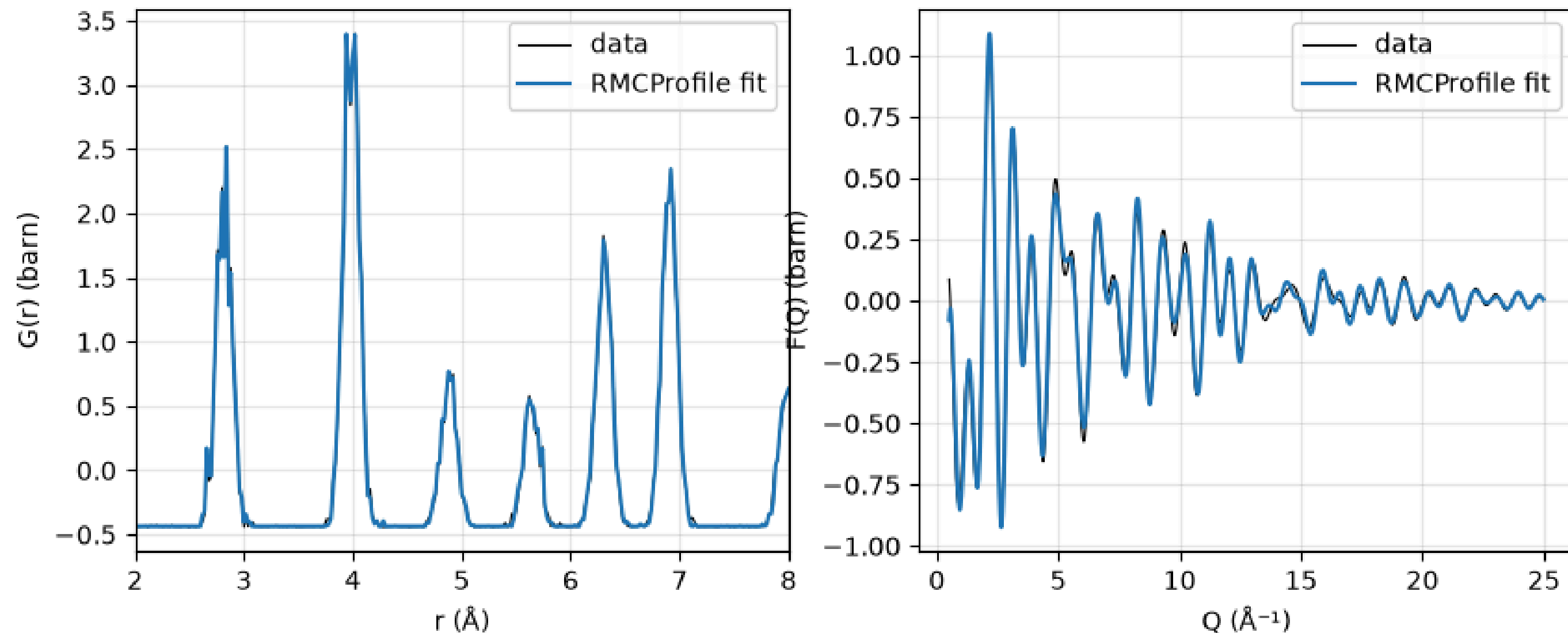
Run the program with TIME_LIMIT 0 on the ideal lattice against the synthetic data: no moves, one evaluation. Its calculated $G(r)$ is the toolkit's, and its experiment column is our data.



Before any move: the synthetic $G(r)$ 'data' (thermal box plus noise) against RMCProfile's calculated $G(r)$ of the ideal-lattice starting box, read back from its _PDF1.csv. The delta-sharp lattice peaks must broaden into the data's. – Figure 7 · notebook §15 · chapter 4, cell 5

The refinement

Half a minute of moves on the CUDA build: tens of thousands of moves, χ^2 from ten thousand to order ten, and the calculated $G(r)$ on the data within the noise.

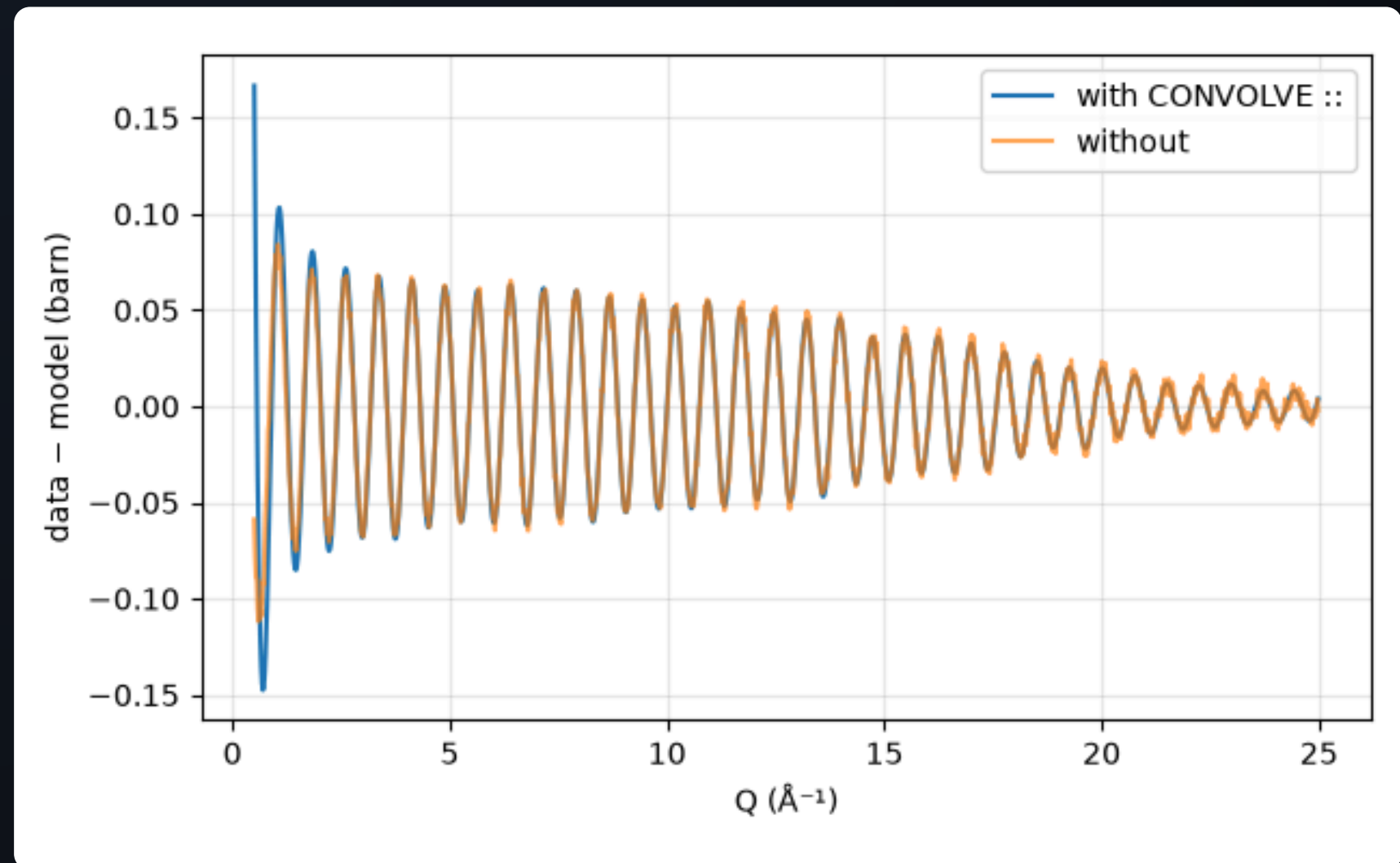


After 30 s of RMCProfile moves on the synthetic NaCl set: the calculated $G(r)$ (left) and $F(Q)$ (right) against the data they were fitted to. – Figure 8 · notebook §16 · chapter 4, cell 7

CONVOLVE :: and the finite box

A reciprocal-space fit against $F(Q)$ must account for the box: RMCProfile convolves its calculated $F(Q)$ with the box function. The keyword's empty value is not optional.

- With CONVOLVE the calculated $F(Q)$ carries the same ringing as data transformed from a finite r range; without it the fit chases the ripples.
- The experiment column of the output is the data **convolved** when CONVOLVE is on — compare like with like.
- A constant offset (CONSTANT_OFFSET) is the other reciprocal-space knob; the notebook compares runs with and without it.



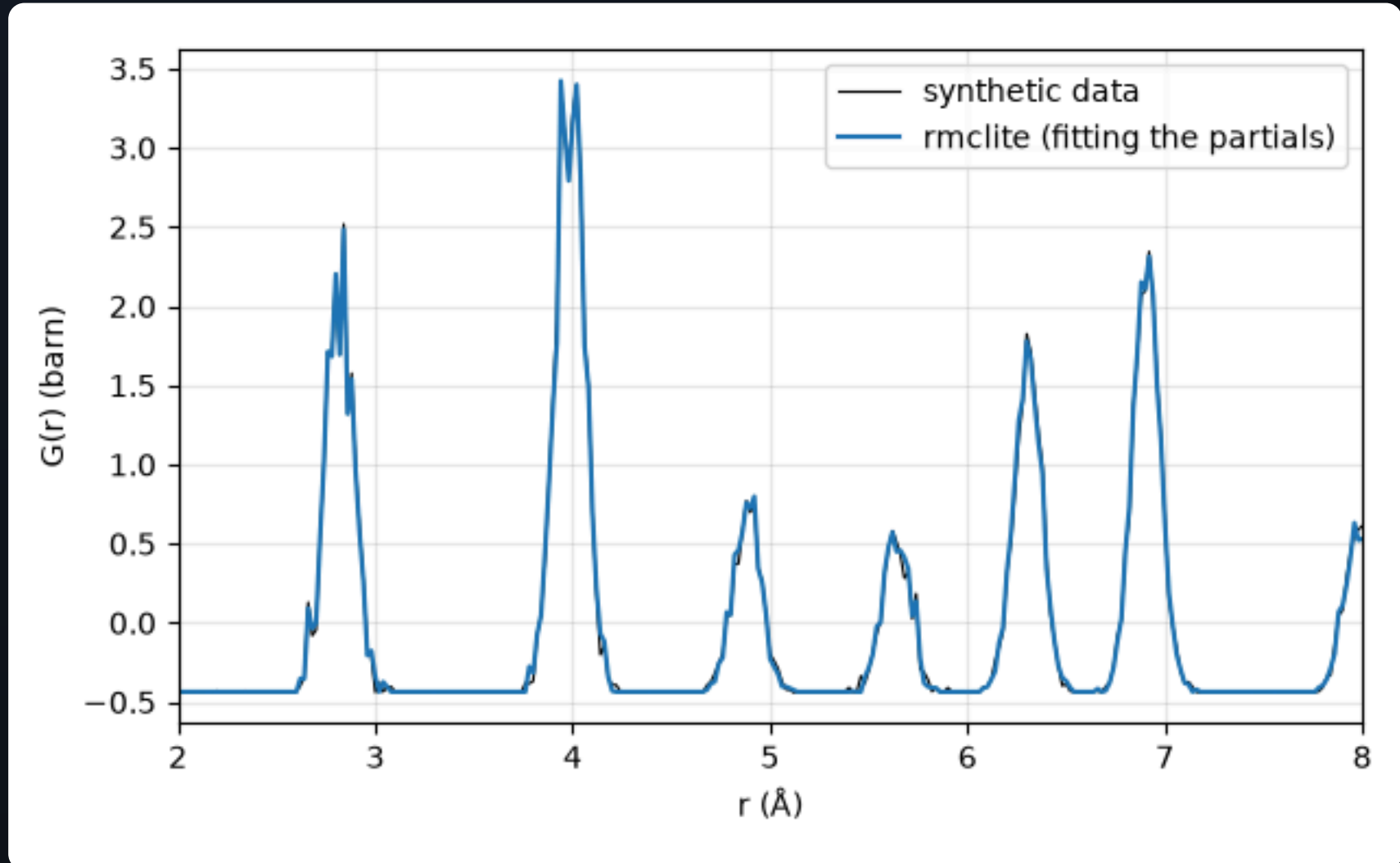
Residual of the reciprocal-space fit of the refined box with and without CONVOLVE ::. Without it the data keep their sharp features while the model, cut at the half box, rings — the residual at high Q grows.

– Figure 9 · notebook §17 · chapter 4, cell 9

The same data through rmclite

Feed the same synthetic $G(r)$ to the notebooks' engine: it converges to the same pair distribution, slower and in the open — every step readable.

- Same grid, same weights, same closest approach: the two engines are compared on their calculated functions, not on their boxes.
- rmclite's fit from the ideal lattice cuts χ^2 by four orders of magnitude on a total $G(r)$ target as it did on partials.
- What differs is speed and features — RMCProfile fits Bragg profiles, EXAFS and magnetic data; rmclite fits pair functions and teaches.



The same synthetic data reproduced by rmclite after 8000 moves from the ideal lattice; rmclite fitted the three partials, RMCProfile in §16 fitted the total $G(r)$ and $F(Q)$.

– Figure 10 · notebook §18 · chapter 4, cell 11

The box function

Truncating the transform at $r_{\max}$ is a multiplication in r and a convolution in Q ; the fit must apply the same operation to the model.

TRUNCATED TRANSFORM

$$F_{\text{box}}(Q) = \int F(Q') M(Q - Q') dQ', \quad M(Q) = (1/\pi) \int_0^{r_{\max}} \cos(Qr) dr = \sin(Q r_{\max}) / (\pi Q)$$

WHAT CONVOLVE DOES

$$F_{\text{calc}} \rightarrow F_{\text{calc}} \otimes M, \quad F_{\text{data}} \rightarrow F_{\text{data}} \otimes M$$

THE FITTED QUANTITY

$$\chi^2_F = \sum_Q [(F_{\text{calc}} \otimes M)(Q) - (F_{\text{data}} \otimes M)(Q)]^2 / \sigma^2$$

- The ripples of lecture 1's figure have period $2\pi/r_{\max}$: $0.74 \text{ \AA}^{-1}$ for an $8.46 \text{ \AA}$ half-box.
- The notebooks do not reproduce RMCProfile's $F(Q)$ column to single precision as they do $G(r)$: the program convolves with its own $r_{\max}$ and grid; an $F(Q)$ cross-check is future work.

Constraints as physics

Closest approach, distance windows, bond potentials, fixed coordination and bond valence sums — measured on refined boxes.

- 1 Why data are not enough
- 2 Closest approach: MINIMUM_DISTANCES
- 3 Windows, potentials, coordination
- 4 Bond valence sums
- 5 The restraint terms

Why data are not enough

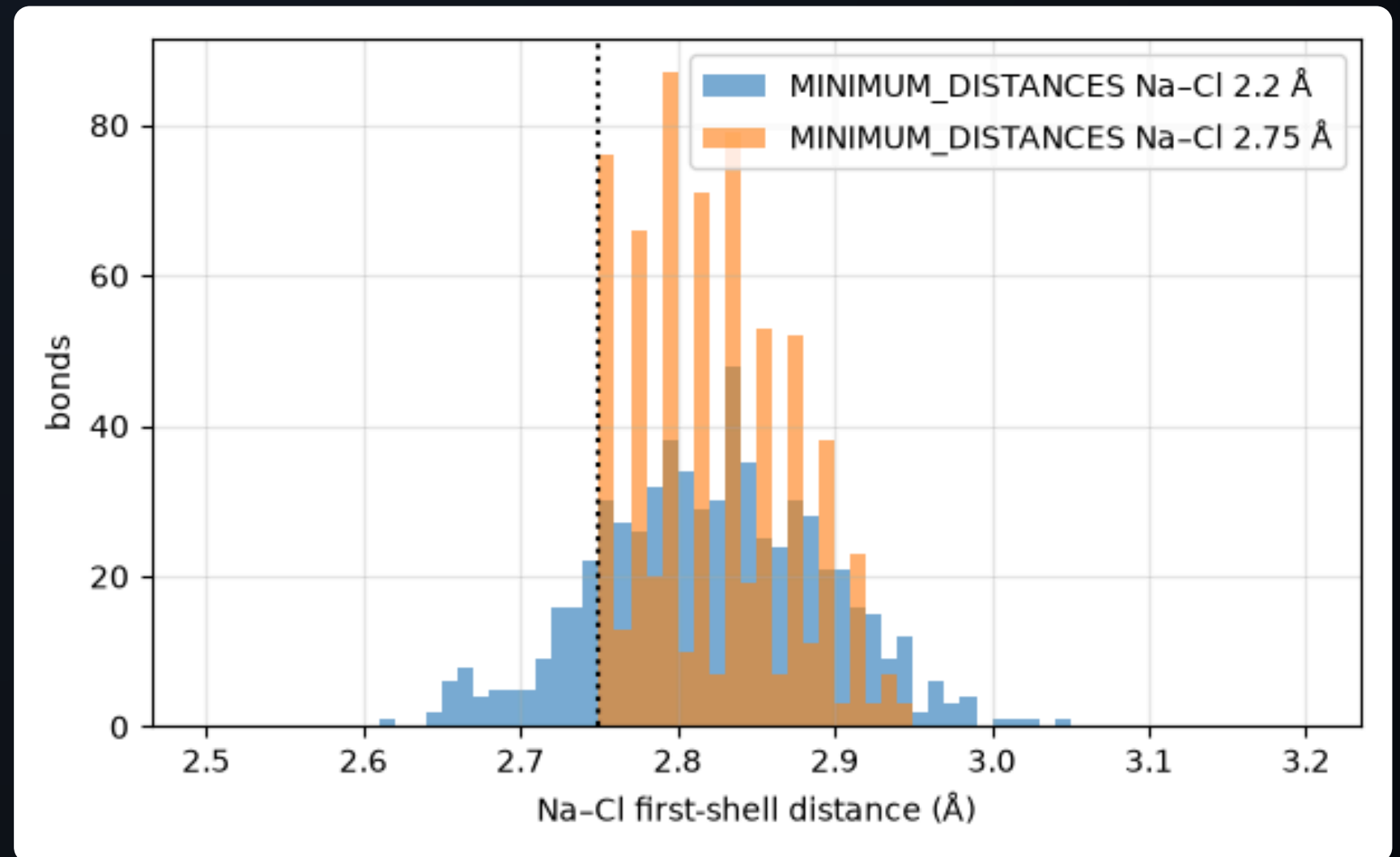
A fit that reproduces $G(r)$ can still put two chlorines where a sodium belongs, break a coordination polyhedron, or let an atom drift through a wall of neighbours. Constraints say what the data cannot.

- Hard constraints make a move impossible (closest approach, distance window, fixed coordination).
- Soft restraints penalise it (bond potentials, bond valence sums) and enter the acceptance rule as an energy.
- Every one of them is a keyword block in the .dat file; lecture 6 runs each for six seconds on the NaCl set and measures the refined box.

Closest approach: `MINIMUM_DISTANCES`

One number per pair below which no two atoms may come. Raise the Na–Cl minimum above the inner edge of the thermal shell and the refined box cannot reach the data's peak from below.

- With the minimum at 2.2 Å the first shell is reproduced on both sides of 2.82 Å; at 2.75 Å it is cut off there and χ^2 stays two orders of magnitude higher.
- The program's message when it stops early — 'Check the `MINIMUM_DISTANCES` section' — is about the **count** of numbers, which the checker verifies first.
- Choose the minimum from chemistry (ionic radii, a known shortest bond), never from the data's noise floor.



The Na–Cl first-shell distances of two refined boxes: with the closest approach at 2.2 Å the thermal shell is reproduced on both sides of 2.82 Å; with it at 2.75 Å (dotted) no bond can be shorter, and the distribution is cut off there. — Figure 15 · notebook §24 · chapter 6, cell 5

Windows, potentials, coordination

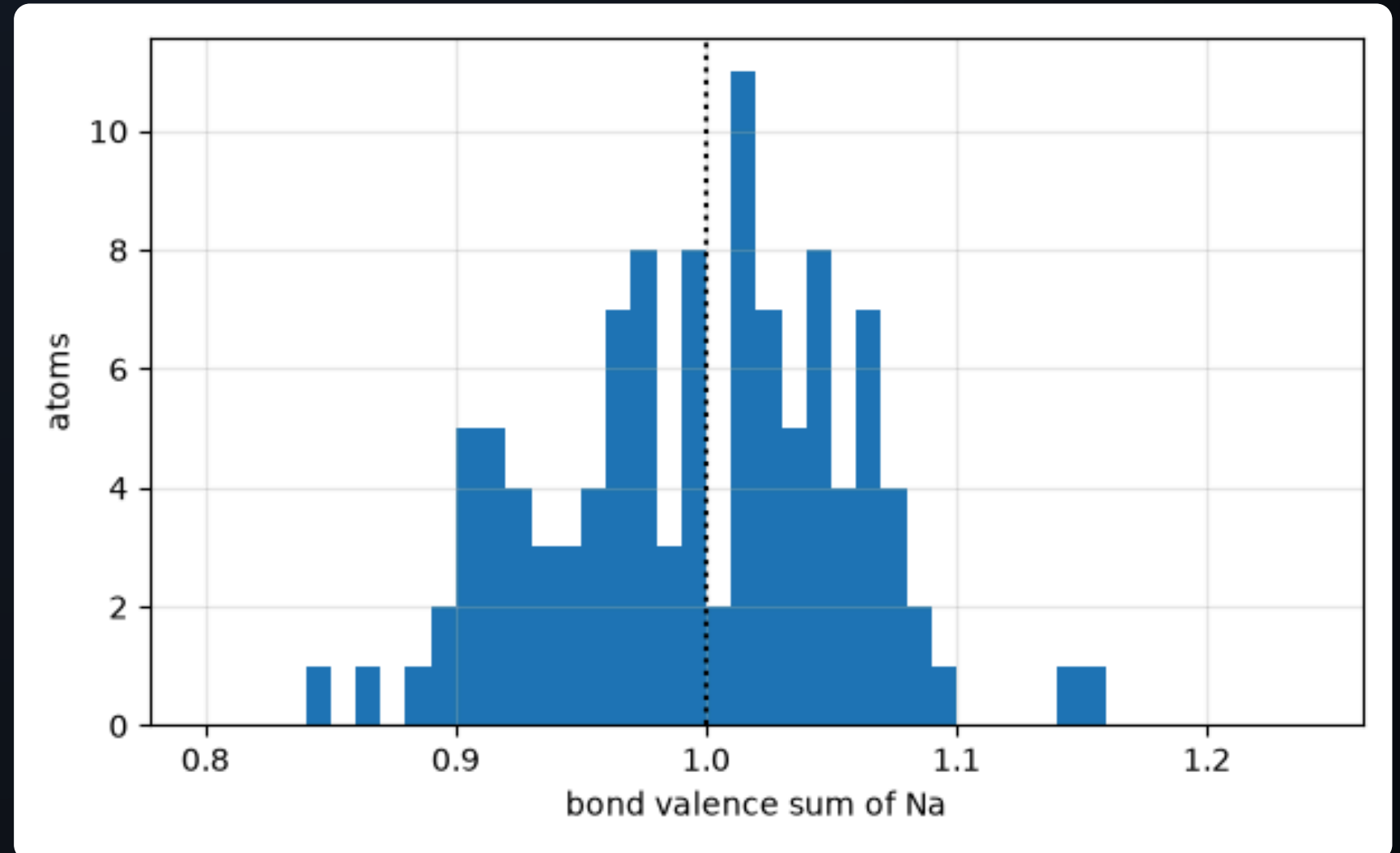
Three ways to keep the neighbours you start with. Chapter 6 runs each for six seconds and measures the box RMCProfile saved.

- `DISTANCE_WINDOW ::` with MNDIST/MXDIST bounds a pair from both sides and freezes the neighbour list at the start (a `.neigh` file): with the window every Na–Cl first-shell distance stays within [2.60, 3.05] Å even under 0.5 Å moves; without it, six-fold coordination is lost.
- `POTENTIALS ::` with STRETCH, STRETCH_SEARCH and TEMPERATURE adds a harmonic bond: a stiff 40 eV/Å² term narrows the Na–Cl shell whatever the data say.
- `FIXED_COORDINATION_CONSTRAINTS ::` asks that a fraction of Na keep six Cl within a distance range; the program reports 100.00 % at every print and the refined box keeps 6 : 108 within 3.2 Å.

Bond valence sums

A bond valence sum turns a coordination shell into a chemical check: it should equal the ion's oxidation state. RMCProfile's `BVS ::` block restrains the fit toward the nominal valences; the toolkit computes the same quantity on any box.

- For Na–Cl with $R_0 = 2.15 \text{ \AA}$ and $b = 0.37 \text{ \AA}$ the ideal lattice gives $V(\text{Na}) = 0.981$.
- The box refined under the BVS restraint has $V(\text{Na}) = 0.995 \pm 0.03$ over 108 atoms, centred on +1 — the figure.
- The chi2 file gains a BVS column: the restraint is one more term in the total the run minimises.



Bond valence sums of the 108 Na atoms in the box refined under RMCProfile's BVS restraint, computed with the toolkit's own calculator ($R_0 = 2.15 \text{ \AA}$, $b = 0.37 \text{ \AA}$, cutoff $3.2 \text{ \AA}$): centred on +1, the nominal valence.

– Figure 16 • notebook §27 • chapter 6, cell 11

The restraint terms

Every constraint on this lecture's slides is one of three expressions added to the acceptance rule.

HARD WALL

$$P_{\text{accept}} = 0 \text{ if } d_{ab} < d_{ij}^{\text{min}} \text{ or } d_{ab} \notin [MNDIST, MXDIST]$$

BOND POTENTIAL

$$\Delta U/k_B T, \quad U = 1/2 k (d - d_o)^2 \text{ for } |d - d_o| \leq \text{STRETCH_SEARCH}$$

BOND VALENCE SUM

$$V_a = \sum_b \exp[(R_o - d_{ab})/b], \quad \chi^2_{BVS} = w \sum_a (V_a - V_{\text{nominal}})^2$$

- The wall costs nothing when it refuses: no histogram update, no χ^2 evaluation.
- STRETCH_SEARCH defines which pairs are bonds at the start — the bond list is generated once, like the neighbour list.
- The BVS weight multiplies a sum over atoms, so it scales with box size; RMCProfile's WEIGHTS entry is per atom type.

Corrections

What the instrument does to $G(r)$ and what RMCProfile does about it — measured, not assumed: the damping is Gaussian, the nanoparticle keyword corrects the baseline.

- 1 What the instrument does to $G(r)$
- 2 Damping, measured: Gaussian, not exponential
- 3 The right damping wins
- 4 Broadening: BROADENING_CORRECTION
- 5 Nano-size: what PARTICLE_RADIUS actually changes
- 6 The three corrections, as the program applies them

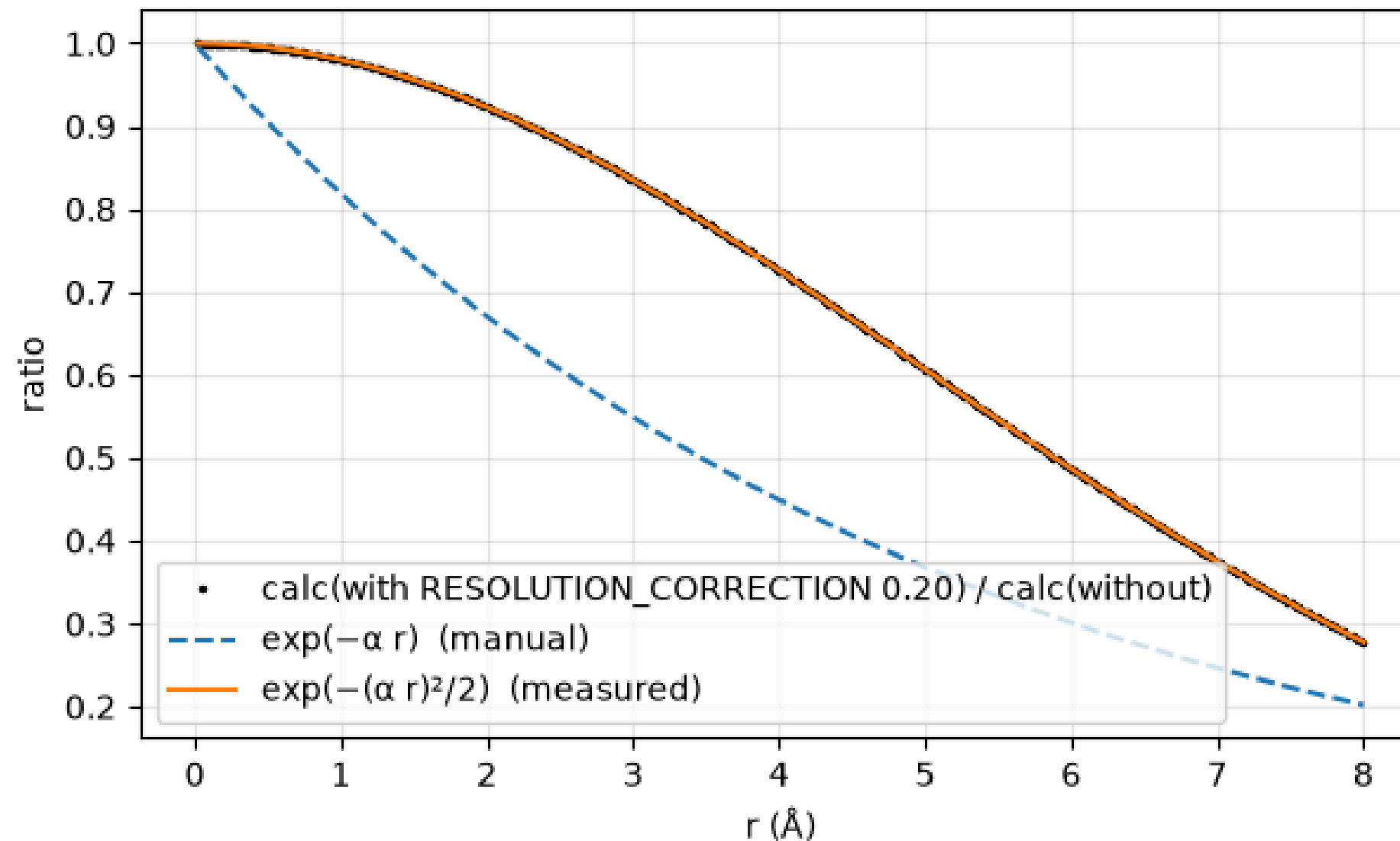
What the instrument does to $G(r)$

A measured $F(Q)$ ends at a finite $Q_{\max}$ and is smeared by the instrument's resolution; its transform is damped at large r and its peaks are broadened. RMCProfile corrects the **calculated** function to look like the data.

- **RESOLUTION_CORRECTION** damps the calculated real-space function with r .
- **BROADENING_CORRECTION** convolves it with a Gaussian whose width grows with r .
- **PARTICLE_RADIUS** (with **BULK_RHO**) applies a nanoparticle shape function; **R_CUTOFF** zeroes the Fourier ripples below the closest approach.
- Lecture 7's rule: before trusting a formula, measure what the program applies.

Damping, measured: Gaussian, not exponential

The manual describes RESOLUTION_CORRECTION as $\exp(-r \cdot \text{Expo})$. The ratio of RMCProfile's calculated columns with and without the keyword is $\exp(-(\alpha r)^2/2)$ to 10^{-3} — PDFgui's Q_{damp} form — and departs from the exponential by 0.25 at 8 Å.



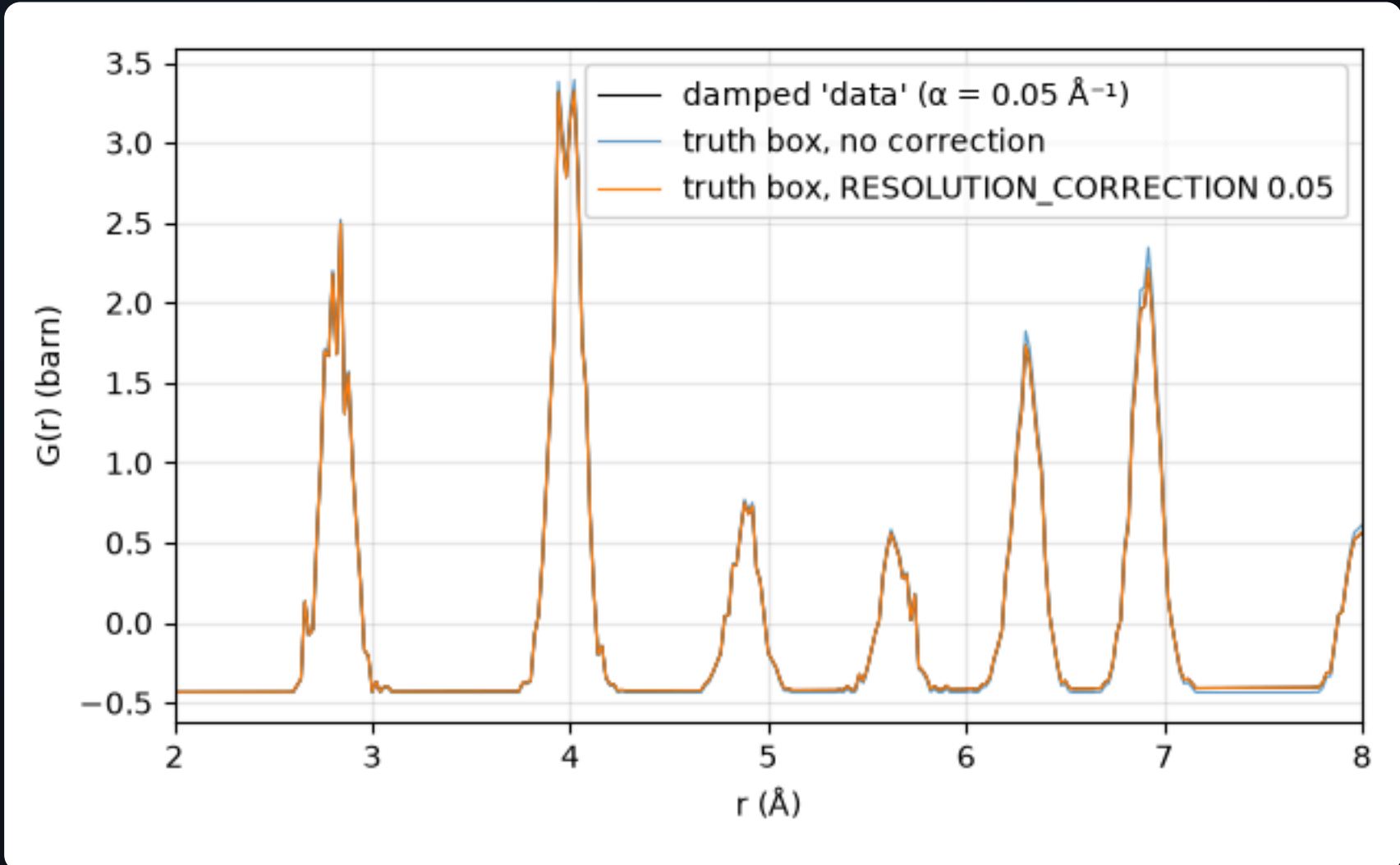
The damping RMCProfile applies for RESOLUTION_CORRECTION 0.20, measured as the ratio of two calculated $G(r)$ columns of the same box: a Gaussian in r , not the exponential the manual writes.

— Figure 17 · notebook §29 · chapter 7, cell 5

The right damping wins

Synthesise data from the truth box's $G(r)$ damped by $\exp(-(0.05 r)^2/2)$. Evaluate the truth box against them with no correction, with $\alpha = 0.05$ and with $\alpha = 0.20$: the right α has the lowest χ^2 .

- The model is the truth box itself, so the only difference between model and data is the correction under test.
- Exercise 7.1 scans α over 0.02, 0.05, 0.10: the minimum sits at 0.05, and the well is not sharp — resolution parameters are best taken from a calibrant, not fitted.
- Convert a PDFgui Q_{damp} directly: it is the same α .

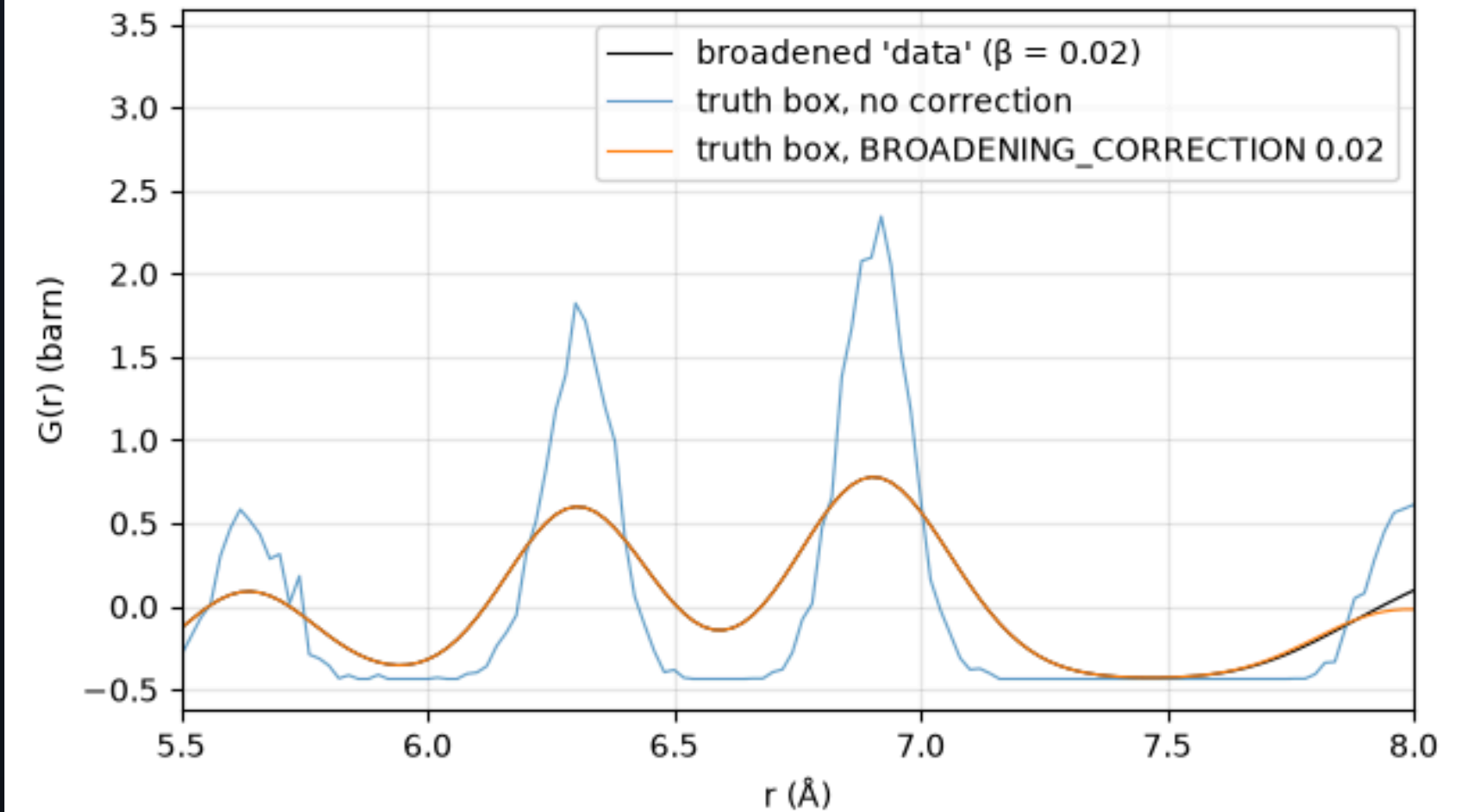


Synthetic $G(r)$ damped by $\exp(-(0.05 r)^2/2)$ against RMCProfile's calculated $G(r)$ of the truth box without and with the matching RESOLUTION_CORRECTION: the corrected model's peaks shrink with r the way the data's do. — Figure 18 · notebook §29 · chapter 7, cell 7

Broadening: **BROADENING_CORRECTION**

Finite Q-resolution broadens each peak in r by an amount growing with r : a Gaussian of $\sigma = \beta r$ (manual Appendix F). Synthetic data broadened that way are fitted best by the same β .

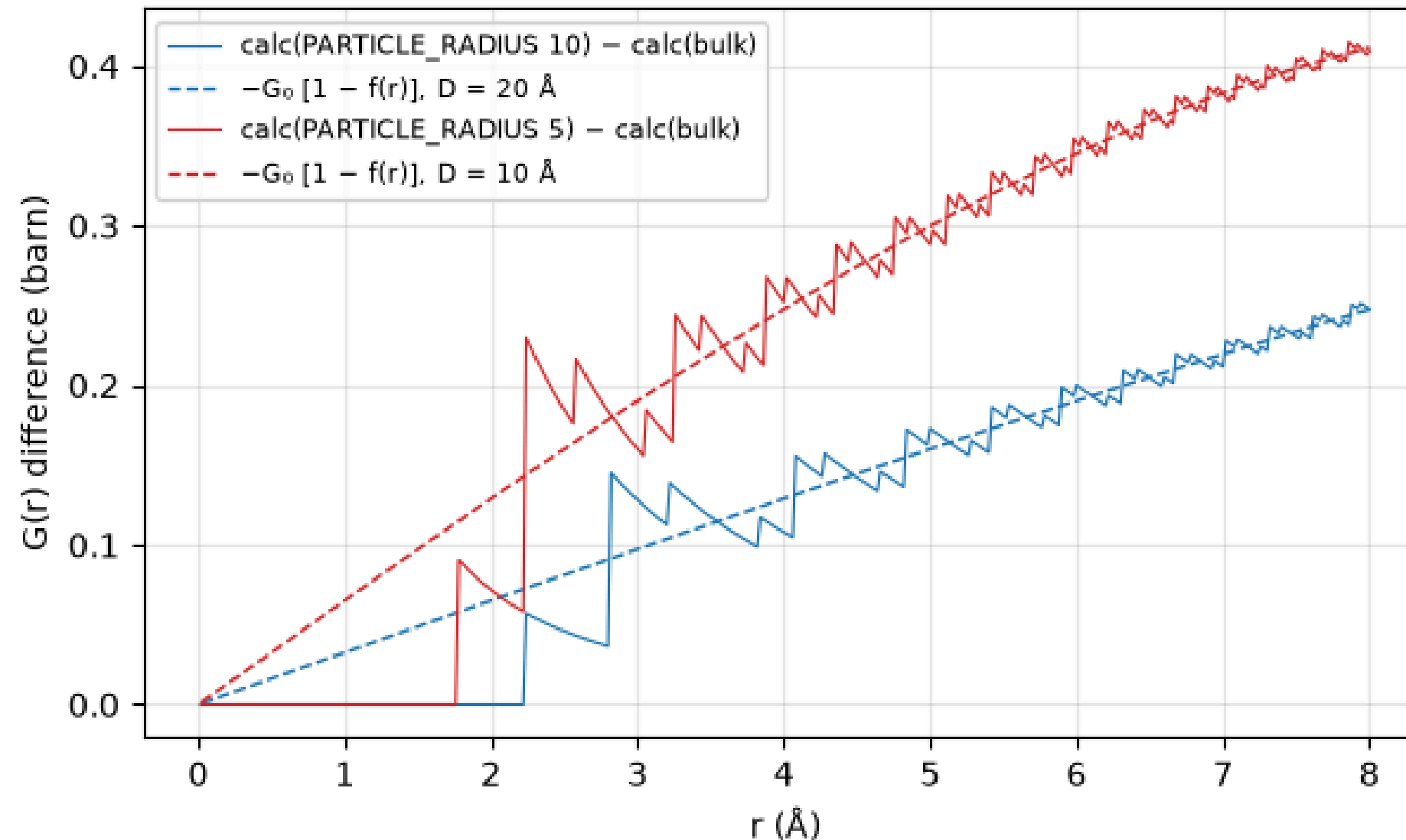
- The toolkit does the r -dependent convolution explicitly; RMCProfile's $\beta = 0.02$ reproduces it and wins the χ^2 comparison by two orders of magnitude against no correction.
- The parameter is not PDFgui's Q_{broad} ; Appendix F relates them.
- At 7.6–8 Å the uncorrected peaks are visibly narrower and taller than the data's.



At large r the r -dependent broadening is visible: the uncorrected truth-box peaks are narrower, the corrected ones spread like the synthetic data's. — Figure 19 · notebook §30 · chapter 7, cell 9

Nano-size: what **PARTICLE_RADIUS** actually changes

Measured the same way, **PARTICLE_RADIUS** does not scale the peaks by a shape function. Beyond the closest approach it adds $-G_0 \cdot [1 - f(r)]$ with f the spherical shape function of diameter $2R$: the bulk baseline replaced by the envelope-weighted one, to 0.01 barn rms.



What **PARTICLE_RADIUS** adds to RMCProfile's calculated $G(r)$ of the same box for radii of 10 and 5 Å (solid), against the bulk baseline $-G_0$ multiplied by one minus the spherical shape function of a particle of diameter $2R$ (dashed): the keyword replaces the uniform background, not the peaks, and does nothing below the closest approach. – Figure 20 · notebook §31 · chapter 7, cell 11

The three corrections, as the program applies them

Each was measured on the same 216-atom box at TIME_LIMIT 0; each is a line a data reduction can be matched to.

DAMPING (RESOLUTION_CORRECTION A)

$$G_{calc}(r) \rightarrow G_{calc}(r) \cdot \exp[-(\alpha r)^2 / 2]$$

BROADENING (BROADENING_CORRECTION B)

$$G_{calc}(r) \rightarrow \int G_{calc}(r') N(r - r' ; \sigma = \beta r) dr'$$

NANO-SIZE (PARTICLE_RADIUS R, BULK_RHO P)

$$G_{calc}(r) \rightarrow G_{calc}(r) - G_o [1 - f(r)], \quad f = 1 - 3r/2D + r^3/2D^3, \quad D = 2R$$

- The damping is the Fourier pair of a Gaussian resolution function in Q of width $1/\alpha$; the manual's $\exp(-r \cdot \text{Expo})$ would correspond to a Lorentzian.
- The Guinier shape function f is the fraction of pair vectors of length r that fit inside a sphere of diameter D : 1 at the origin, 5/16 at $D/2$, 0 at D .

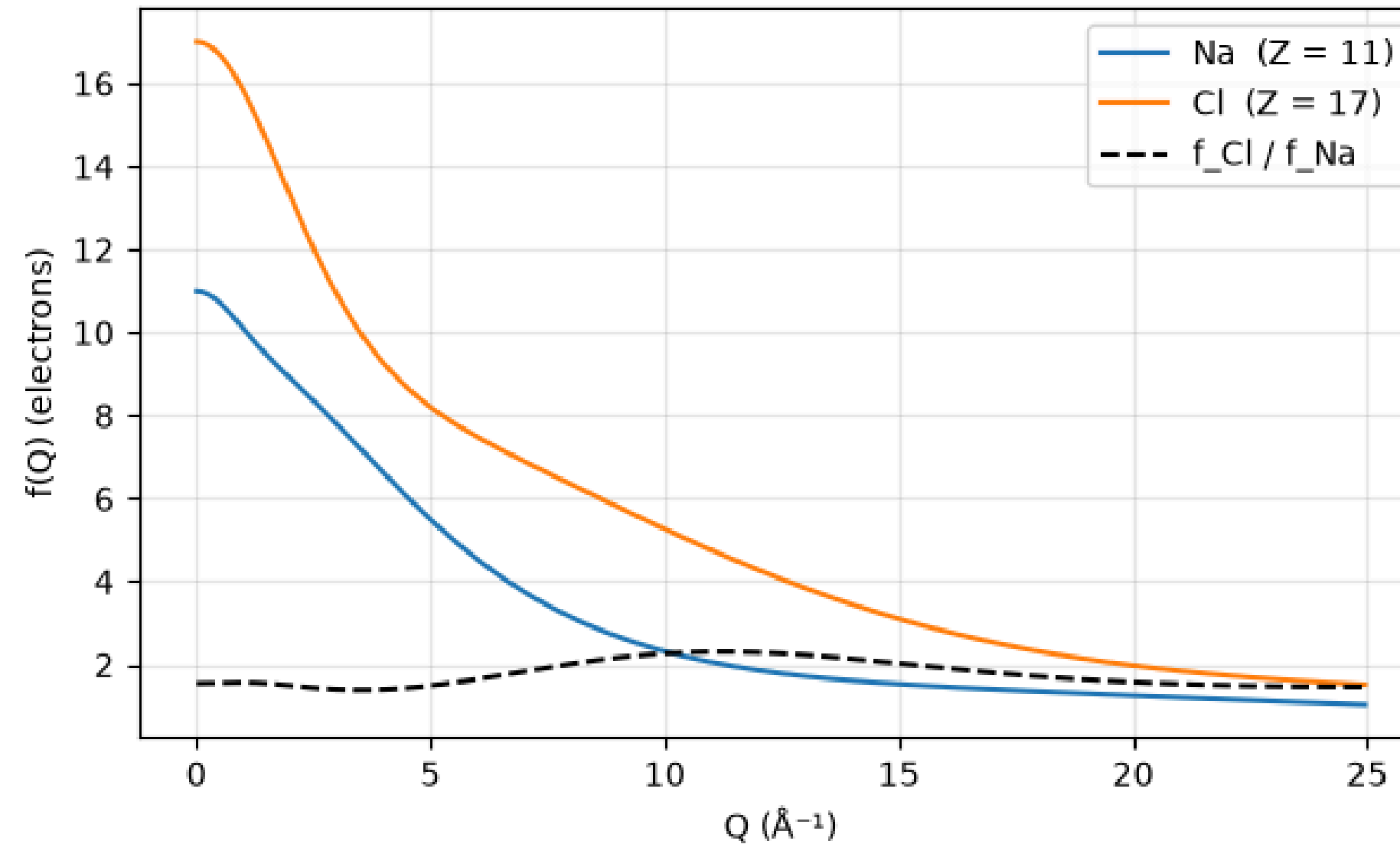
Beyond neutrons

X-ray form factors and contrast, an honest X-ray run, Bragg profiles from a Rietveld refinement, EXAFS on the shipped exercise, magnetic and diffuse fits described.

- 1 X-rays see electrons: the form factor
- 2 Contrast: the same box, two radiations
- 3 An X-ray run, and what it does not confirm
- 4 Bragg profiles: the long-range order
- 5 EXAFS on the shipped exercise
- 6 Magnetic and diffuse: described, not run
- 7 X-ray weights and the normalisation question

X-rays see electrons: the form factor

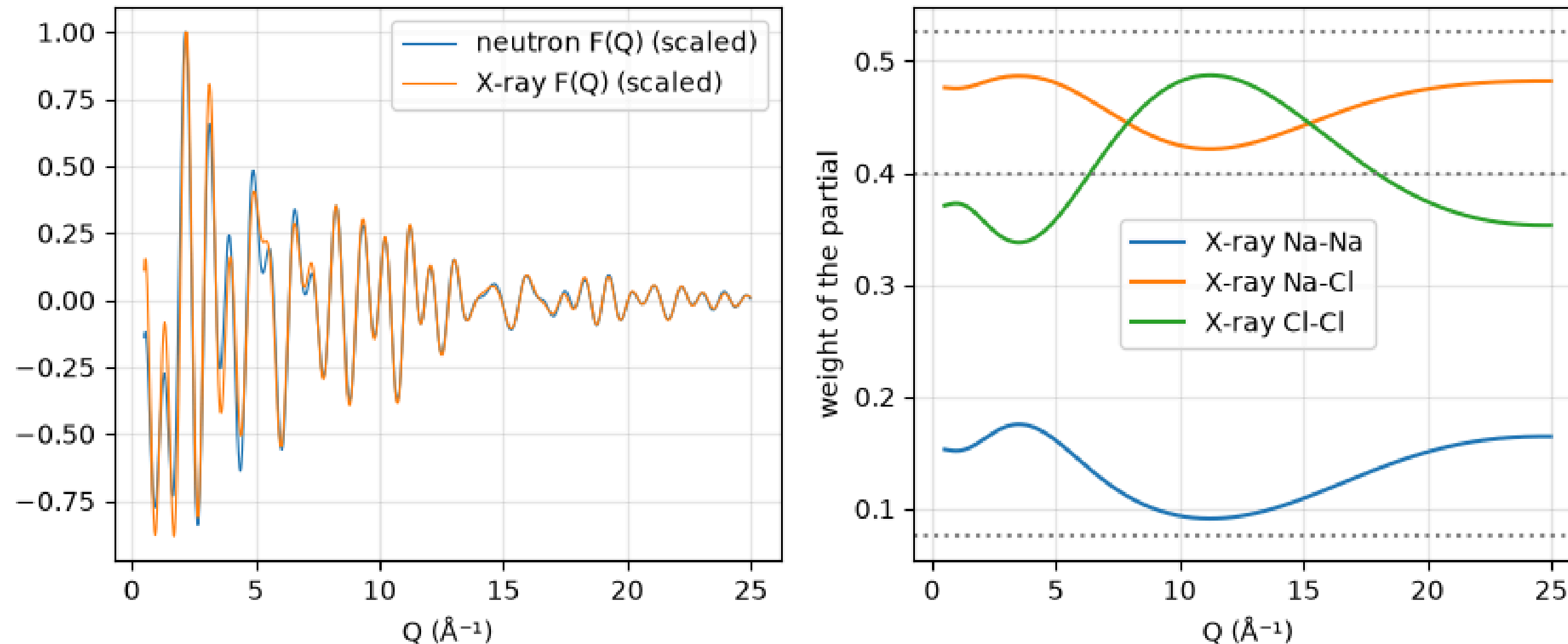
A neutron's scattering length is a constant per isotope; an X-ray's is the atomic form factor $f(Q)$: Z electrons at $Q = 0$, falling with Q . The ratio between two elements is not constant, so an X-ray $F(Q)$ mixes the partials with Q -dependent weights.



X-ray atomic form factors of Na and Cl from the Cromer–Mann tables: Z electrons at $Q = 0$, falling with Q ; the ratio between them is not constant, so an X-ray $F(Q)$ mixes the partials with Q -dependent weights. – Figure 11 · notebook §19 · chapter 5, cell 3

Contrast: the same box, two radiations

Neutrons weigh Cl–Cl most (3.63 versus 9.58 fm); X-rays weigh Na–Cl more and the balance drifts with Q . Two data sets on one box see different partials — the reason to fit both.

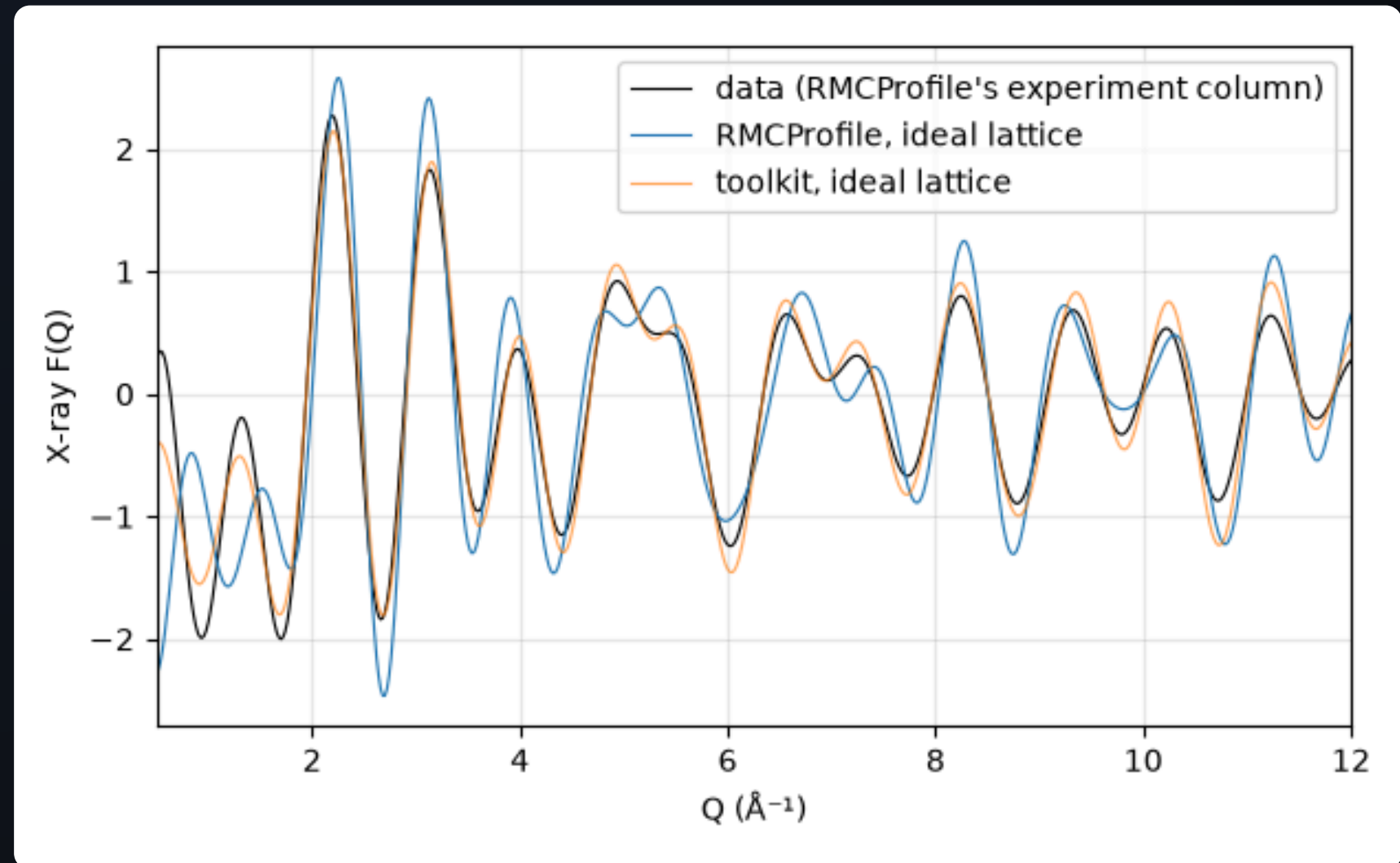


Left: the neutron and X-ray total scattering functions of the same thermal NaCl box. Right: the X-ray weights of the three partials along Q (solid) and the neutron ones (dotted, normalised): X-rays weigh Cl–Cl less and Na–Cl more than neutrons do, and the balance drifts with Q . – Figure 12 · notebook §20 · chapter 5, cell 5

An X-ray run, and what it does not confirm

The X-ray block runs, reports χ^2 , returns our data exactly in its experiment column, and its calculated column has our scale — median ratio 0.999 — but its shape agrees with ours only to a correlation of 0.90, and neither NORMALIZATION_TYPE value changes it.

- The neutron functions are reproduced to single precision; the X-ray $F(Q)$ processing is not — finding N-8, a question for the mailing list, not a lowered threshold.
- What the notebook claims is exactly what it measured: same scale, same peak positions, partial agreement in shape.
- The .xray file the toolkit writes is accepted by the program; the checker validates the block.

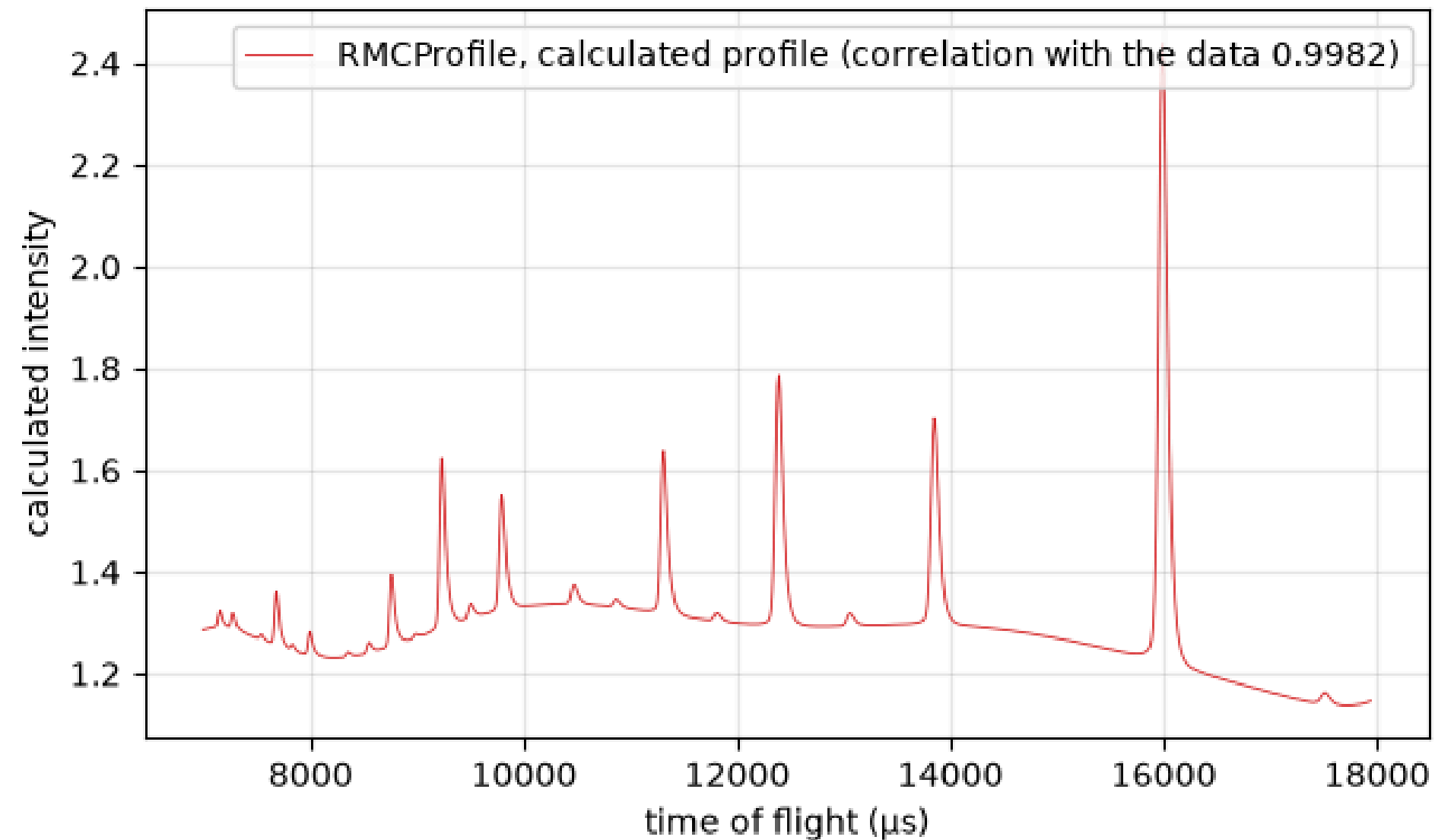


RMCProfile's X-ray $F(Q)$ of the ideal NaCl lattice against the toolkit's for the same box and the synthetic data: same scale and peak positions, a shape that agrees only in part — the open question of finding N-8.

— Figure 13 • notebook §21 • chapter 5, cell 7

Bragg profiles: the long-range order

RMCPProfile is the RMC code that fits the Bragg profile together with the PDF. The BRAGG block needs four files from a Rietveld program; on the shipped SF₆ exercise the toolkit reads them, the program evaluates the profile, and the shipped configuration reproduces it with correlation 0.998.

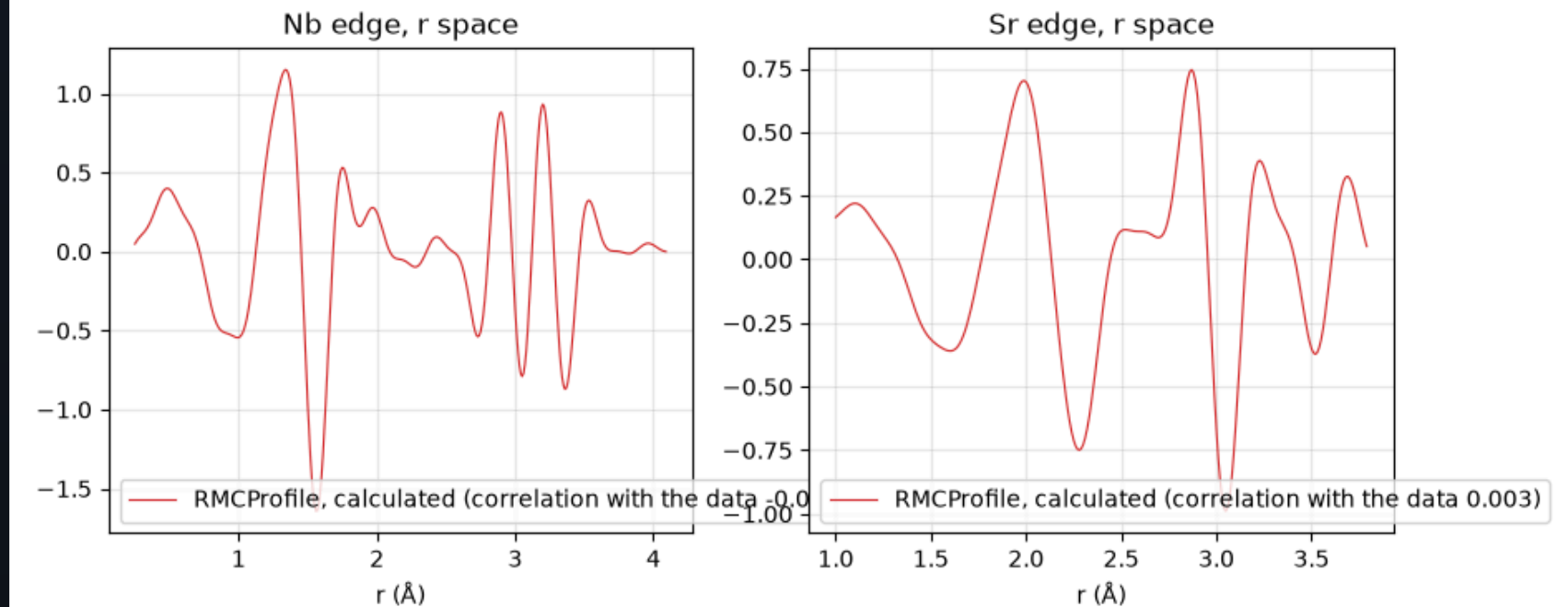


RMCPProfile's calculated Bragg profile for the shipped SF₆ configuration — the average structure of the 4 320-atom box, fitted together with $G(r)$ and $F(Q)$. The exercise's measured profile is not reproduced here (it belongs to the package); its agreement with this curve is the correlation printed in the legend. — Figure 14 · notebook §22 · chapter 5, cell 9

EXAFS on the shipped exercise

The EXAFS block fits the absorption fine structure $\chi(k)$ of one absorbing element, computed from the box with FEFF-style path amplitudes. The shipped SnO exercise evaluates two edges in eleven seconds at TIME_LIMIT 0 and writes k-space and r-space outputs per edge.

- Keywords per edge: FILENAME, TYPE(S)_OF_ABSORBING_ATOMS, START/END_POINT in k and in r, FIT_space, R_SPACING, K_POWER, ENERGY_OFFSET, WEIGHT.
- The k-space start and end points are k values in $\text{\AA}^{-1}$ (3.3–13.2 in the exercise), not point indices as in the PDF blocks — the exercise reads it off the output.
- The absorber and scatterer lists and the path files come from the exercise's preparation step, not from RMCProfile.



The two EXAFS edges of the package's SnO exercise (Nb and Sr absorbers) in r space: RMCProfile's calculation for the shipped starting configuration, evaluated with no moves. The exercise's measured $\chi(r)$ is not reproduced here (it belongs to the package); the legend gives the calculation's correlation with it.

– Figure 21 · notebook §32 · chapter 8, cell 3

Magnetic and diffuse: described, not run

Two data types this course cannot reproduce with its own engine and the package ships no exercise for: the notebooks document their keyword blocks so a reader with such data knows what the program expects.

- `MAGNETISM ::` — `FORM_FACTOR`, `MAGNETISM_FILE_STEM`, `MAGNETIC_ATOMS`, `MAX_SPIN_MOVEMENT`, `SPIN_MOVE_RATE`: spins refined alongside positions against the magnetic pair distribution (Paddison & Goodwin 2012 for the idea).
- `DIFFUSE_SCATTERING3D ::` — a measured reciprocal-space volume against the box's, with `SUPERCELL_DIMENSIONS` declared so the average amplitude can be computed; costly per move.
- Where this leaves the course: every pair-distribution data type reproduced, X-rays reproduced in scale and partly in shape, Bragg and EXAFS run on the shipped exercises, magnetic and diffuse described from the manual.

X-ray weights and the normalisation question

The three partials, two sets of weights; and the two conventions a data reduction must match.

TOOLKIT'S X-RAY $F(Q)$

$$F_X(Q) = \sum_{ij} c_i c_j f_i(Q) f_j(Q) [S_{ij}(Q) - 1] / (\sum_i c_i f_i(Q))^2$$

THE OTHER CONVENTION

$$\dots / \sum_i c_i f_i(Q)^2 \quad (NORMALIZATION_TYPE :: \langle f^2 \rangle, \text{the default})$$

CROMER-MANN FORM

$$f(s) = \sum_{k=1..4} a_k \exp(-b_k s^2) + c, \quad s = Q / 4\pi$$

- Both normalisations make the weights sum to one at every Q in the $(\sum cf)^2$ case and nearly so in the other; they differ by a few per cent along Q .
- The measured RMCProfile column matches neither in shape (correlation 0.90 against both), while matching the $(\sum cf)^2$ scale (ratio 0.999) — so the difference is not the normalisation.
- The Appendix-D route treats X-ray $G(r)$ as neutron data with $b \rightarrow Z$: constant weights, an approximation the toolkit also provides.

Reading a configuration

A refined box is a statistical object: coordination numbers, bond lengths, bond angles, displacement clouds and ensembles.

- 1 A refined box is a statistical object
- 2 Coordination numbers and bond lengths
- 3 Bond angles and displacement clouds
- 4 Ensembles: several fits are better than one
- 5 Distributions from a box

A refined box is a statistical object

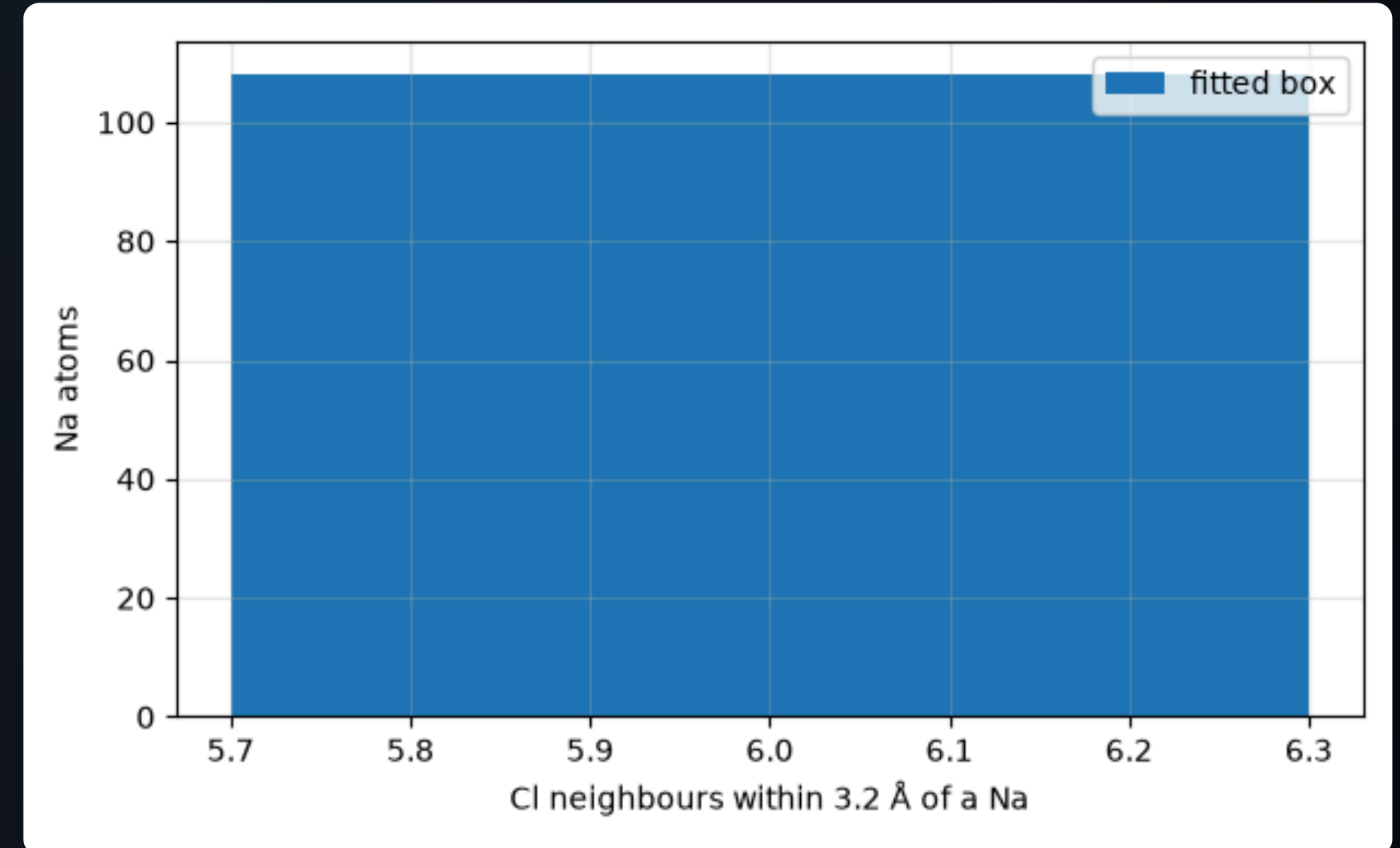
What one reads off a box must be distributions — coordination numbers, bond lengths, bond angles, displacement clouds — never the position of a particular atom. Four analyses, each checked against the answer the synthetic truth knows.

- `coordination(cfg, A, B, rmax)` : per-atom counts and their histogram.
- The bond-length distribution is the partial itself, weighted by the shell: its mean and width are the two numbers a PDF is usually asked for.
- `bond_angles(cfg, A, B, C, rmax)` : every B–A–C angle whose arms are within rmax.
- Displacement of each atom from its own ideal site, projected — the displacement cloud.

Coordination numbers and bond lengths

In rock salt every Na keeps six Cl within 3.2 Å whatever the thermal motion: the histogram is a single bar, 6 : 108. The Na–Cl bond from the fitted box is 2.822 ± 0.074 Å against the truth's 2.822 ± 0.072 Å.

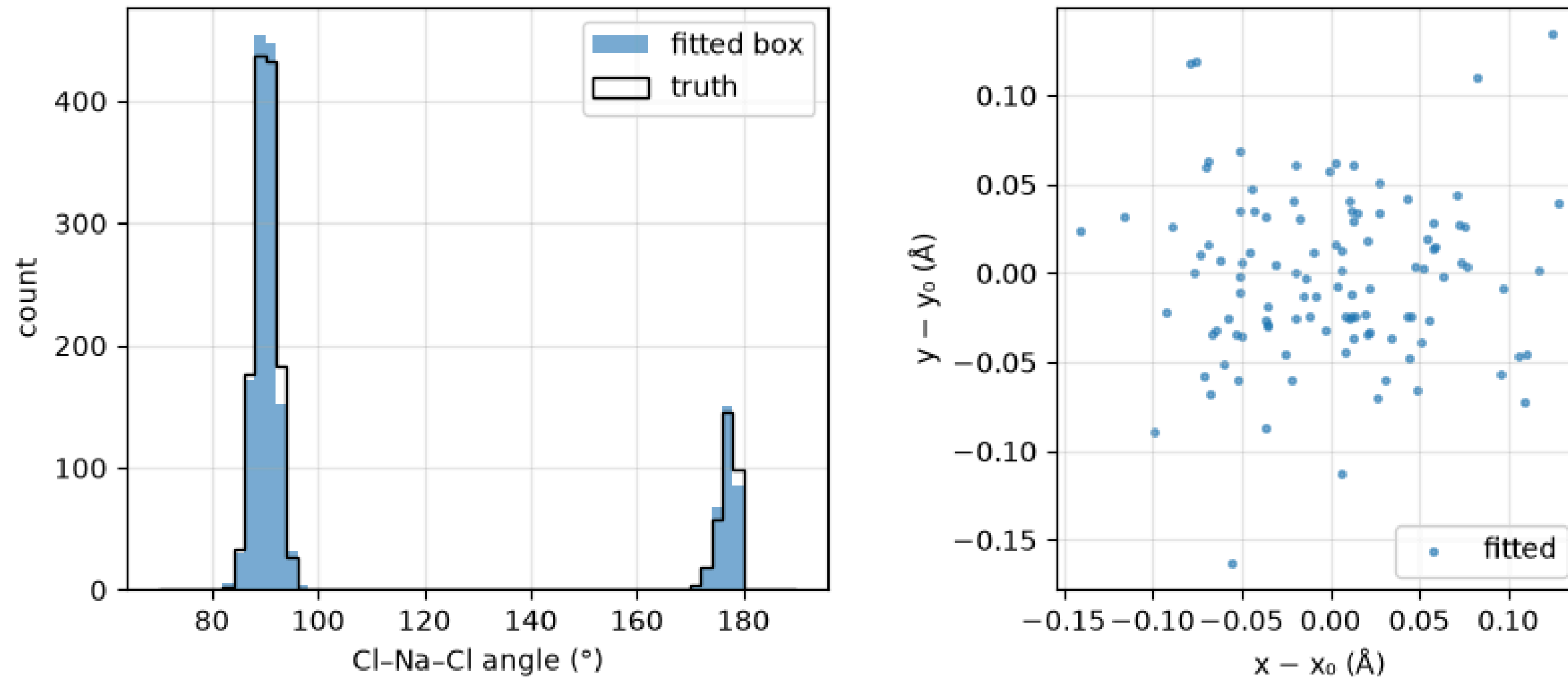
- Mean and width come from the Na–Cl partial weighted by r^2 over the first shell, the same for fit and truth.
- Coordination does not change in a close-packed ionic crystal; thermal motion broadens shells, it does not move atoms between them.
- The second shell too: twelve Na–Na neighbours within 4.3 Å for every Na (exercise 9.1).



Coordination-number histogram of the refined NaCl box: every one of the 108 Na atoms keeps its six Cl neighbours — thermal motion broadens shells, it does not change coordination in a close-packed ionic crystal. — Figure 22 · notebook §36 · chapter 9, cell 5

Bond angles and displacement clouds

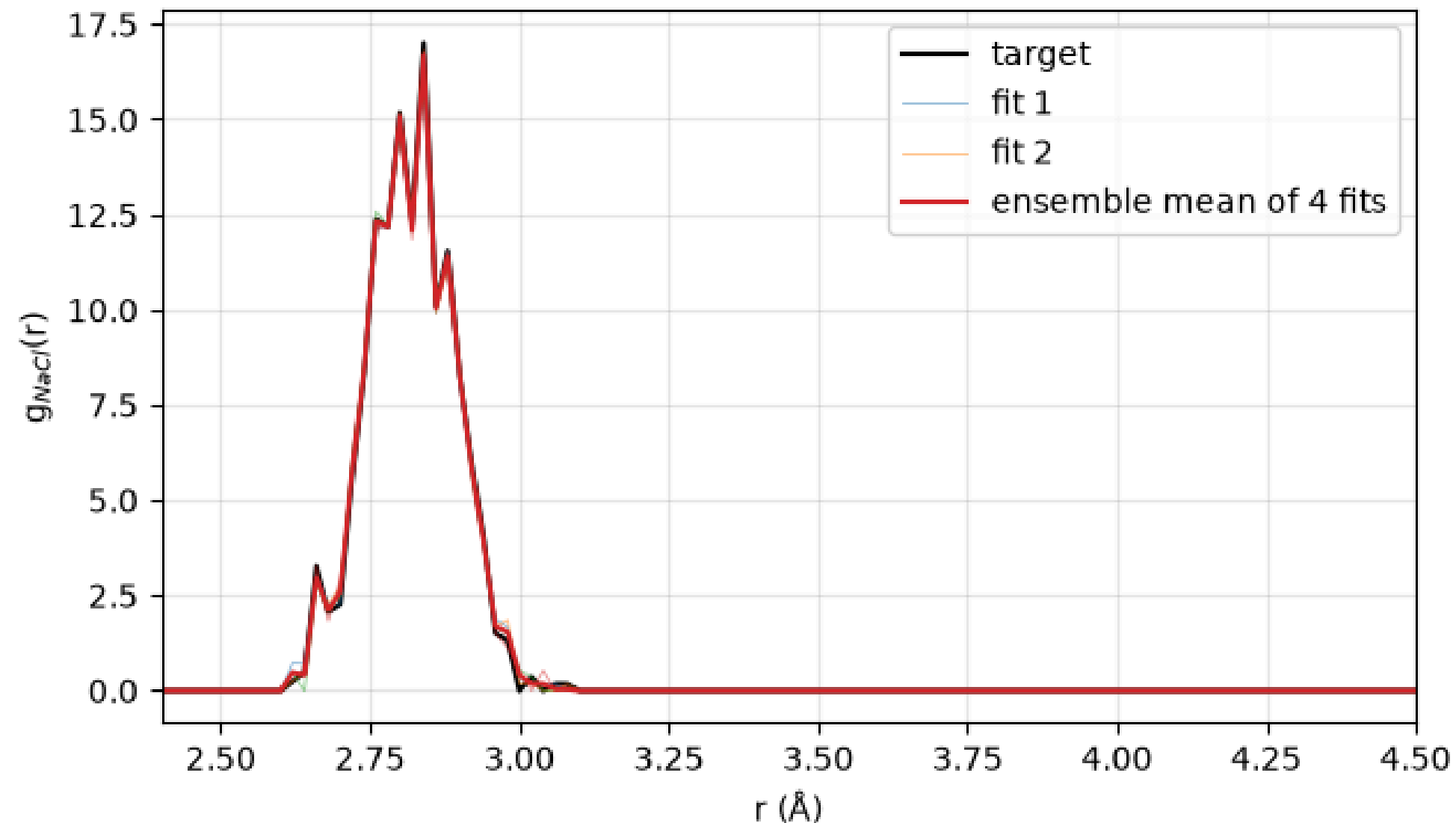
Cl–Na–Cl angles peak at 90° and 180° , broadened by the thermal displacements; the fit's 90° peak sits at 89.9° with a spread of 2.08° against the truth's 2.02° . The right panel is every Na's displacement from its own ideal site — the cloud a displacement ellipsoid summarises.



Left: the Cl–Na–Cl angle distribution of the refined box (filled) and of the truth box (line): two peaks at 90° and 180° broadened by the thermal displacements. Right: the displacement of each of the 108 Na atoms from its ideal site, projected on *ab* — the displacement cloud of the Na site. — Figure 23 · notebook §37 · chapter 9, cell 7

Ensembles: several fits are better than one

One fitted box carries the statistical noise of its 216 atoms. Four fits from different seeds, averaged, are closer to the target than any single fit — and the spread between them is the honest error bar of anything read off a box.



Four rmclite fits of the same synthetic data from different seeds (thin) and their mean (red) against the target (black): averaging over an ensemble beats any single box, and the spread between boxes is the error bar. — Figure 24 · notebook §38 · chapter 9, cell 9

Distributions from a box

Every number on the previous slides is one of these moments.

BOND LENGTH FROM THE PARTIAL

$$\langle d \rangle = \Sigma_{shell} r_k^3 g(r_k) / \Sigma_{shell} r_k^2 g(r_k), \quad \sigma_d^2 = \Sigma r_k^2 g(r_k) (r_k - \langle d \rangle)^2 / \Sigma r_k^2 g(r_k)$$

DISPLACEMENT FROM THE IDEAL SITE

$$u_a = [x_a - x_a^{ideal} - \text{round}(x_a - x_a^{ideal})] \cdot L$$

ENSEMBLE ESTIMATE

$$\bar{g} = (1/N) \Sigma_n g_n, \quad \text{rms}(\bar{g} - g_{target}) < \min_n \text{rms}(g_n - g_{target})$$

- The r^2 weighting turns $g(r)$ back into a pair count per shell before averaging.
- The $\text{round}()$ is the minimum-image wrap in fractional coordinates; L the lattice matrix.
- Independent fits have independent noise; averaging N of them reduces it as $1/\sqrt{N}$ until the systematic residual is reached.

Contributing to RMCProfile

The live cross-check and what it costs, the ecosystem around RMCProfile, and how findings become reports the developers can use.

- 1 The cross-check, live, and what the numbers cost
- 2 The ecosystem

- 3 How findings become contributions
- 4 Measure, don't assume: the ratio method

The cross-check, live, and what the numbers cost

Everything the notebooks compute on their own is compared with the package's output on the package's own exercise: partials equal to 2.8×10^{-4} (its single precision), $G(r)$ to 3×10^{-5} barn, on both builds, against recorded tolerances that were measured, not chosen.

- `upstream_adapter` stages a shipped exercise in a scratch directory, runs the program and compares; the records file stores tolerances, measured maxima and a provenance line — never the package's data.
- The smoke test reproduces the exercise's χ^2/dof of 0.4201 on Windows (CUDA) and on Linux (CPU); the deterministic passes print identical numbers on both.
- Cost: `rmclite` ~1 900 moves/s on 64 atoms; `RMCProfile` ~20 000 moves/s with $G(r)$ only and far fewer with an $F(Q)$ set — the Fourier transform per move is what reciprocal space costs.

The ecosystem

RMCPProfile's site lists the neighbours it considers relevant; one honest line each, no ranking.

tool	what it is	relation to RMCPProfile
RMC++ (Budapest)	the general RMC code descended from RMCA, for liquids and glasses	the common ancestor; RMCPProfile added crystals, Bragg profiles and the constraint families
HRMC (CSIRO)	hybrid RMC with interatomic potentials	the idea RMCPProfile's POTENTIALS block adopts for specific bonds
fullrmc	a Python RMC engine with a modular constraint system	a scriptable alternative; no Bragg-profile fitting
EPSR / Dissolve (ISIS)	empirical potential structure refinement for liquids and glasses	data-driven potentials rather than direct moves
DISCUS	diffuse-scattering simulation and refinement of disordered crystals	the rmc_to_discus export; another route to the same disorder
PDFgui / diffpy-CMI	small-box PDF refinement and its scriptable successor	the average-structure-plus-parameters description RMC's big box complements
GudrunN/X, PDFgetX3, Mantid, ADDIE	data reduction from raw scattering to F(Q) and G(r)	what produces the files the notebooks synthesised
Topas4RMC, sofq_calib, rmc_tools	RMCPProfile's own Python side tools (conda, GPL)	the surface where an outside contributor's patches can go

How findings become contributions

RMCProfile is closed-source, so contributions are reports, reproductions and documentation — through its mailing list, under the contributor's own name. The study behind this course keeps a ledger; every entry has a run log, a file and a line.

- PROVEN findings from the shipped exercises: a missing input file stops the program with exit code 0 (P-11); a missing .poly makes it wait forever (P-12); a data file's letter case breaks the Linux build (P-10).
- Measured against the manual: the damping is Gaussian (P-17); PARTICLE_RADIUS needs BULK_RHO (P-18) — each a chapter check and a draft note to the developers.
- Open: the X-ray $F(Q)$ processing (N-8) — a question, drafted, never sent by a machine.
- Patches go to the GPL Python tools, with tests; nothing from the package is ever redistributed.

Measure, don't assume: the ratio method

The technique behind lecture 7's findings generalises: two deterministic evaluations of the same box, one keyword apart, isolate what the keyword does.

MULTIPLICATIVE KEYWORD

$$R(r) = G_{calc}^{with}(r) / G_{calc}^{without}(r) \text{ away from zero crossings}$$

ADDITIVE KEYWORD

$$\Delta(r) = G_{calc}^{with}(r) - G_{calc}^{without}(r)$$

THE VERDICT

$$\max | R - \exp(-(ar)^2/2) | < 10^{-3}, \quad \text{rms} | \Delta + G_o(1 - f) | < 0.02 \text{ barn}$$

- TIME_LIMIT 0 makes both evaluations deterministic; the truth box as model makes the columns smooth enough to divide.
- A hypothesis is a function to compare with; the number reported is the deviation, with its threshold stated before the run.
- The same method cross-checks the toolkit against the program: our function, its column, the maximum difference, recorded at the measured precision.

Where to go next

The notebooks are the course; the program's manual and tutorials are the reference; the mailing list is the community.

- Run the eleven chapters on your own machine; change the truth box (a different salt, a different displacement) and see which checks still pass.
- Install RMCProfile 6.7.9 under its own terms and run the shipped exercises through the toolkit's cross-check; then your own data.
- Read Tucker et al. 2007 (RMCProfile) and Keen 2001 (the functions); Playford et al. 2014 for a review of what RMC has found in crystals.
- When something in the manual and the program disagree, measure it, write it down, and tell the developers.